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AI-Driven Decision Support for Warehouse Operations: A Qualitative Case Study of Decision-Making in Practice

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Abstract

This thesis how practitioners perceive AI-driven decision support systems to influence operational performance, resilience and sustainability in their warehouse operations, and identifies the organizational and technical factors that determine their effective use. As warehouses encounter increasing complexity due to volatile demand, multi-supplier coordination, and frequent disruptions, traditional planning methods relying on human judgment, statistical forecasting, and rule-based systems exhibit notable limitations. A qualitative case study design is adopted, guided by an interpretivist research philosophy. Empirical data are collected through semi-structured interviews with practitioners and solution providers from multiple organizations, supplemented by observations from the Plan Konferensen 2026 industry conference. The data are analyzed using the Gioia methodology, resulting in a three-level coding structure comprising first-order concepts, second-order themes, and aggregate dimensions.

The findings are structured around five aggregate dimensions: Human-AI Coordination, AI-Supported Planning and Replenishment, Operational Value Creation, Implementation Barriers and Enabling Factors, and Emerging AI Capabilities. The results indicate that AI primarily serves as an augmentation tool rather than as an autonomous decision-maker, shifting planners' responsibilities from routine calculations to exception management and validation. Operational benefits include enhanced service levels, reduced excess inventory, improved lead time predictability, and more scalable replenishment planning. Resilience is increased through greater disruption visibility and faster exception response. Sustainability benefits, such as waste reduction and transport consolidation, arise indirectly from improved planning efficiency. Effective AI utilization depends on a set of technical preconditions, such as master data quality, integration of warehouse management systems (WMS) and enterprise resource planning (ERP) systems, and real-time data availability, as well as organizational factors, including user trust,

explainability, change management, training and governance. The study concludes that AI-supported warehouse replenishment should be understood as a socio-technical decision support process in which value is created through the interaction of analytical systems, human judgment, and organizational context, rather than through complete automation.

Keywords

Artificial Intelligence, Decision Support Systems, Warehouse Replenishment, Supply Chain Management, Human-AI Coordination, Operational Performance, Sustainability, Resilience, Augmentation, Automation

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AI-drivet beslutsstöd för lagerverksamhet: En kvalitativ fallstudie av beslutsfattande i praktiken

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Sammanfattning

Detta examensarbete undersöker hur AI-drivna beslutsstödsystem påverkar operativ prestation, resiliens och hållbarhet inom lagerpåfyllnad, samt vilka organisatoriska och tekniska faktorer som är viktiga för att systemen ska användas effektivt. I takt med att lagerverksamheter blir allt mer komplexa, exempelvis genom varierande efterfrågan, samordning mellan flera leverantörer och återkommande störningar, blir begränsningarna i traditionella planeringsmetoder tydligare. Dessa metoder bygger ofta på mänskliga bedömningar, statistiska prognoser och regelbaserade system.

Studien genomförs som en kvalitativ fallstudie med en interpretivistisk forskningsansats. Empirin samlas in genom semistrukturerade intervjuer med praktiker och lösningsleverantörer från flera organisationer samt genom observationer från branschkonferensen Plan Konferensen 2026. Datamaterialet analyseras med hjälp av Gioia-metodologin, där analysen struktureras i tre steg: första ordningens koncept, andra ordningens teman och aggregerade dimensioner.

Resultaten presenteras utifrån fem övergripande dimensioner: samspel mellan människa och AI, AI-stödd planering och lagerpåfyllnad, operativt värdeskapande, implementeringshinder och möjliggörande faktorer samt framväxande AI-förmågor. Studien visar att AI främst används som ett stöd i beslutsfattandet snarare än som en helt självständig beslutsfattare. Planerarnas arbete förändras från att utföra rutinmässiga beräkningar till att i större utsträckning hantera avvikelser och validera systemets rekommendationer.

De operativa fördelarna omfattar förbättrade servicenivåer, minskat överskottslager, bättre förutsägbarhet i ledtider och mer skalbar planering av lagerpåfyllnad. Resiliensen stärks genom bättre överblick vid störningar och snabbare hantering av avvikelser. Hållbarhetsfördelar, såsom minskat svinn och effektivare transporter, uppstår främst indirekt genom förbättrad planering.

Studien visar också att effektiv användning av AI kräver flera tekniska förutsättningar, exempelvis hög kvalitet på masterdata, integration mellan lagerhanteringssystem (WMS) och affärssystem (ERP) samt tillgång till realtidsdata. Samtidigt är organisatoriska faktorer viktiga, såsom

användarnas tillit till systemen, förklarbarhet, förändringsledning, utbildning och tydlig styrning. Slutsatsen är att AI-stödd lagerpåfyllnad bör förstås som en socioteknisk beslutsstödsprocess där värde skapas genom samspelet mellan AI-system, mänskligt omdöme och den organisatoriska kontexten, snarare än genom fullständig automatisering.

Nyckelord

Artificiell Intelligens, Beslutsstödsystem, Lagerpåfyllnad, Försörjningskedjehantering, Människa-AI-Samverkan, Operativ Prestation, Hållbarhet, Resiliens, Augmentering, Automatisering

Table of Contents

1 Introduction 1

1.1 Background 1

1.2 Problem Formulation 3

1.3 Purpose 4

1.4 Research Question 5

1.5 Delimitations 5

1.6 AI Usage 6

2 Literature Review 7

2.1 Supply Chain and Warehouse Management 7

2.2 AI as Decision Support in Supply Chains 8

2.3 AI Mechanisms: Predictive, Prescriptive, Real-Time Adaptation 9

2.4 Limitations of Traditional Approaches 10

2.5 Warehouses as Complex Decision Environments 11

2.6 AI-Supported Replenishment in Practice 12

3 Methodology 17

3.1 Research design 17

3.2 Data collection 18

3.3 Data analysis approach 22

3.4 Research quality and limitations 25

3.5 Ethical Considerations 27

4 Findings 28

4.1 Human-AI Coordination 28

4.1.1 AI as a Decision Support Tool Rather Than a Decision-Maker 28

4.1.2 The Shift from Routine Work to Exception Management 29

4.1.3 Complexity and the Continued Indispensability of Human Judgment 29

4.1.4 Uneven Trust and the Need for Explainability 30

4.2 AI-Supported Planning and Replenishment 31

4.2.1 Lead time uncertainty as a primary AI application 31

4.2.2 Generating replenishment proposals by combining various data 32

4.2.3 Exception flagging as a planning support mechanism 32

4.2.4 Predictive slotting and warehouse layout optimization 33

4.2.5 Data quality as a foundational condition 34

4.2.6 Replenishment logic 34

4.3 Operational Value Creation 35

4.3.1 Service level improvements and inventory optimization 35

4.3.2 Waste reduction and sustainability outcomes 36

4.3.3 Order quantity optimization and improvements 37
4.3.4 Workload redistribution and workforce implications 38
4.4 Implementation Barriers and Enabling Factors 39
4.4.1 Technical integration as a primary implementation barrier 39
4.4.2 Change management and resistance to adoption 40
4.4.3 Organizational size, governance, and competitive pressure 41
4.5 Emerging and Expanding AI Capabilities 42
4.5.1 Robotic AI for autonomous order picking 43
4.5.2 Expanding AI capabilities within WMS 44
4.5.3 Expanding AI ecosystem strategy 45
5 Discussion 47
5.1 AI as Augmentation Rather Than Automation 47
5.2 AI-Supported Adaptive Planning for Warehousing 49
5.3 Operational Value: Performance, Resilience, and Sustainability 51
5.4 Preconditions: Data, Integration, Trust, and Organizational Readiness 53
5.5 From Decision Support Toward Automation and Orchestration 55
5.6 Summary of Discussion 57
6 Conclusion 59
6.1 Answer to the Research Questions 59
6.1.1 Answer to RQ1: Influence on Operational Performance, Resilience, and Sustainability 59
6.1.2 Answer to RQ2: Organizational and Technical Factors Shaping Effective Use 60
6.2 Theoretical Contribution 61
6.3 Managerial Implications 62
6.4 Limitations 63
6.5 Future Research 64
7 References 66
8 Appendix 69
8.1 Appendix: Interview Guide 69
8.1.1 Interview Questions (Semi-Structured): Users 69
8.1.2 Interview Questions (Semi-Structured): Solution Providers 70
8.2 Coding Table 71
8.3 Gioia Matrix 77



Table of Figures

Table 1: Table showing the list of interviewees who participated in this study 21
Table 2: Table representing one interview's coding into First-Order Concepts 23
Table 3: Table representing the iteration leading to Second-Order Themes and Aggregate Dimensions 24
Table 4: Table summarizing the First-Order Concepts, Second-Order Themes and Aggregate Dimensions 25

1 Introduction

AI-driven decision support is increasingly redefining and enhancing how supply chains are planned and controlled, especially in nodes where multiple material flows converge and are redistributed. In environments characterized by volatile demand, short product life cycles and frequent disruptions, only relying on human judgment or traditional rule-based systems can lead to suboptimal inventory decisions, higher risk exposure and missed opportunities for performance improvement (Perera et al., 2019). As data volume and complexity increase, AI offers potential to enhance human decision-making by generating more accurate forecasts, optimizing inventory levels, and prioritizing actions across a network of interconnected facilities (Toorajipour et al., 2019).

Within this broader supply chain context, warehouses play a particularly critical role. A warehouse can be understood as a centralized node that consolidates, stores, and redistributes materials or products to downstream locations. In other words, it acts as an interface between suppliers, production sites, and customer-facing locations such as retail stores or last-mile warehouses (Shah & Goh, 2006). Because warehouses coordinate inbound and outbound flows, buffer variability, and support responsiveness to customers, the quality of the decisions taken at this level has a major impact on service levels, lead times, and working capital.

Positioning AI as a decision support tool in warehouses therefore creates a promising setting to study how advanced analytics and machine learning can augment planners, improve operational performance and strengthen resilience while still relying on human expertise for oversight and contextual judgment.

1.1 Background

Supply chain management is increasingly challenged by volatile demand, frequent disruptions, and the accumulation of data that surpasses the capabilities of traditional planning methods. Warehouses, which serve as central coordination nodes for managing multi-supplier inventory, distribution, and replenishment, exemplify these complications and require simultaneous optimization of service levels, costs, and responsiveness (León et al., 2024).

Classic methods, including human judgment, statistical forecasting, and rule-based systems, encounter major limitations in addressing these complications:

- Human judgment is susceptible to biases, specifically in volatile environments. (Perera et al., 2019)
- Statistical methods often assume stable relationships, which are infrequently present in dynamic supply chain contexts. (Yu et al., 2019)
- Rule-based systems lack the ability needed to effectively handle dynamic supply chain conditions. (Gasperis & Facchini, 2025)

AI-driven decision support systems deliver a promising solution by processing complex datasets to generate predictive forecasts, prescriptive recommendations, and real-time adaptations. This study reviews the implementation of these systems in warehouses operations and evaluates their effects on performance, resilience, and sustainability.

While AI-driven decision support can be applied across many parts of the supply chain, its potential impact is mainly shown in settings where multiple material flows converge and where planning decisions have system-wide consequences. In such environments, decisions related to forecasting, inventory positioning, and allocation must be made under high uncertainty and time pressure, with direct implications for service levels, costs, and risk exposure.

The focus will be on replenishment decisions to the warehouse; in other words, decisions about when and how much to order from suppliers in the goal of maintaining proper stock levels for an optimal service level.

Warehouses are centralized nodes within supply chain networks that consolidate inbound flows from multiple suppliers and redistribute materials or products to downstream locations such as production sites or customer-facing locations. By acting as interfaces between upstream and downstream actors, warehouses play a key role in coordinating material and information flows and supporting responsiveness to customer demand. (Shah & Goh, 2006)

The importance of warehouses has increased as supply chains have become more global, fragmented, and exposed to demand volatility and external disruptions. Through inventory pooling and centralized coordination, warehouses can improve service levels and working capital

efficiency. At the same time, this centralization makes warehouses critical leverage points, as decisions taken at the warehouse strongly influence lead times, inventory performance, and risk exposure across the network. (Christopher, 2016)

From an operational perspective, warehouses must balance competing objectives such as service level targets, inventory holding costs, capacity constraints, and responsiveness under uncertainty. This makes warehouses particularly complex in decision environments and highlights the importance of effective planning and decision support in these settings.

Warehouse replenishment decisions involve determining the timing and quantity of orders from suppliers to maintain ideal stock levels and ensure consistent service performance.

Warehouse replenishment is a critical planning activity within the warehouse management, linking demand forecasting, supplier performance, inventory availability and operational implementation. (Peng et al., 2022) Decisions regarding timing and quantity of replenishment affect not only stock levels, but also downstream service performance, working capital, transport efficiency, and waste. (De Jesus Pacheco et al., 2023) The complexity of these decisions increases as warehouses manage extensive SKU portfolios, face uncertain supplier lead times, experience volatile demand and face capacity constraints, among other challenges. (Sharma et al., 2020) Therefore, replenishment serves as a relevant context for investigating the role of AI-driven decision support in operational planning.

AI-driven decision support offers potential means to enhance warehouse replenishment by transforming operational data into forecasts, recommendations, alerts, or suggested actions. However, since replenishment decisions are integrated within established warehouse processes and information systems, the practical use of such tools requires further empirical investigation.

1.2 Problem Formulation

Although interest in AI-based decision support for supply chain management is increasing, empirical understanding of the implementation and use of these systems in warehouses remains limited. Furthermore, the impact of such systems on operational performance, resilience, and sustainability is not well understood. Warehouses are becoming essential coordination nodes, requiring planners to make complex forecasting, replenishment, and allocation decisions under

significant uncertainty and time constraints. (Shekarabi et al., 2025) Traditional planning methods, including human judgement, statistical forecasting, and rule-based systems, regularly fail to address this complexity. (Ahn et al., 2024) Existing research on AI in supply chains primarily emphasizes algorithmic performance, with less attention to everyday usage, organizational context, and the changing responsibilities of human planners. Therefore, it is necessary to examine how AI-driven decision support is integrated into a warehouse decision process. To be more precise, this analysis will prioritize replenishment decisions, including determining when to trigger orders from suppliers, how much to order for each SKU, and how to adjust order quantities in response to variations in demand, lead times, or other disruptions.

Current research on AI decision support for warehouse replenishment indicates a maturing field. Systematic reviews cover AI in inventory broadly, while specific models focus on optimizing replenishment in uncertain conditions. However, gaps persist in empirical evidence regarding operational outcomes and real-world barriers such as WMS integration and user skepticism, echoing interview data that show implementation lags potential. The generic AI decision-support literature spotlights the need for explainable, data-efficient systems in changing contexts motivating both, the impacts of AI in warehouses and its factors.

Jinjin Li et al. (2025) report a significant increase in publications on this topic between 2021 and 2025, indicating an increase in research interest in the application of artificial intelligence (AI) in warehouse operations. However, Bergsma et al. (2025) contend that while AI shows promise in inventory management, the supporting evidence remains fragmented and is frequently derived from simplified scenarios that do not accurately represent real warehouse environments. Notably, a substantial gap persists between strong algorithmic effectiveness and practical use, particularly in multi-item systems, stochastic lead times, and other operational complications relevant to replenishment decisions.

1.3 Purpose

The purpose of this study is to investigate how AI-driven decision support systems can be applied in warehouses to improve inventory performance, enhance the resilience of supply chains, and support more sustainable operations; while clarifying how humans interact with and act upon AI-generated recommendations.

More specifically, the study aims to explore how predictive, prescriptive, and real-time adaptive AI mechanisms can be used in a warehouse context to support replenishment decisions to the warehouse. In addition to that, the study also intends to identify organizational and technical factors that enable or hinder their effective use by planners.

1.4 Research Question

The study is guided by the following comprehensive research question:
RQ1: How does the implementation of AI-driven decision support for warehouse replenishment influence operational performance, resilience, and sustainability?
RQ2: What organizational and technical factors shape its effective use?

This question reflects the dual ambition of the thesis; first, to understand the performance implications of AI-supported decisions at a critical node of the supply chain, and second, to explore the conditions under which such systems are successfully adopted in practice. To operationalize this, the study focuses on AI-driven decision support for warehouse replenishment decisions; examining how AI tools are used to plan and adjust orders to suppliers, how these uses affect inventory levels, service levels, lead times, and risk exposure, and how planners interact with AI recommendations.

1.5 Delimitations

This study focuses on the use of AI-driven decision support in warehouses, particularly how such tools support operational performance, risk resilience, and sustainability in warehouses. The study is limited to the warehouse context and does not aim to analyze end-to-end supply chain design or manufacturing planning decisions. Furthermore, this study does not evaluate the technical performance of AI algorithms in terms of model development or computational accuracy but instead investigates how AI is implemented and used in practice, and how it influences operational outcomes.

The study does not include simulation modelling or the development of new AI solutions. Sustainability and resilience are treated as potential outcomes of AI-driven decision support rather than as main research topics, and the study therefore does not include detailed environmental

accounting such as lifecycle assessment. Due to time and access constraints, the study focuses on a limited number of cases and does not aim for statistical generalization.

1.6 AI Usage

This thesis makes controlled use of AI-based tools as a support for the research process, without delegating core academic work to automated systems. AI will primarily be used to assist in identifying relevant literature, organizing references, and refining the formulation parts of the writing. All retrieved sources, concepts and formulations suggested by AI tools are manually verified, assessed, and properly referenced by the authors to ensure accuracy, traceability, and academic integrity via academic search platforms like Google Scholar. Additionally, AI-based translation support was used to translate the abstract from English to Swedish.

2 Literature Review

2.1 Supply Chain and Warehouse Management

Supply chain management involves coordinating material, information, and financial flows across organizations and functions to align supply with demand efficiently and responsively. Instead of examining individual firms in isolation, supply chain management points out the interdependence among suppliers, manufacturers, warehouses, distribution centers, retailers, and customers. Therefore, operational performance depends not only on local efficiency but also on the capacity to coordinate decisions across the broader supply chain network. (Chopra & Meindl, 2022)

Within this broader system, warehouses operate as key physical and informational nodes that link upstream supply to downstream demand. While commonly associated with storage and handling, warehouses also support planning and coordination by buffering uncertainty, consolidating flows, ensuring inventory availability, and enabling responsiveness to variations in demand or supply. As supply chains become more global and susceptible to disruptions, warehouses increasingly act as decision points where inventory, replenishment, and allocation choices impact performance across the entire network. (Christopher, 2016)

Warehouse management acts as an important operational component of supply chain management. Decisions at the warehouse level influence service levels, stock availability, lead times, inventory costs, and working capital. For instance, replenishment decisions determine the timing, quantity, and positioning of inventory to meet anticipated demand. These are not solely internal concerns, as inadequate replenishment planning can result in downstream stockouts, excess inventory, delivery delays, or waste. Thus, warehouse management links operational execution to broader supply chain objectives, including availability, cost efficiency, resilience, and sustainability. (Chopra & Meindl, 2022)

Replenishment is especially significant because it integrates forecasting, inventory control, supplier performance, and warehouse availability. In centralized warehouse settings, replenishment decisions typically involve multiple products, suppliers, lead times, and downstream demand locations. This complexity demands planners to balance objectives such as maintaining service levels, minimizing inventory holding costs, managing uncertainty, and

preventing obsolete or unsellable stock. These factors make replenishment a pertinent area for examining AI-driven decision support, as AI tools can assist planners by enhancing forecasts, identifying risks, recommending order quantities, and enabling more adaptive planning decisions. (Toorajipour et al., 2021)

Consequently, AI-supported warehouse replenishment should be regarded not only as a warehouse management tool but also as a form of supply chain decision support. The warehouse establishes the operational context, while replenishment decisions link this context to broader supply chain performance. A 2020 study shows that advanced decision support models enable organizations to analyze and select resilience strategies in supply chains, underscoring how AI-based tools in warehouses can affect both inventory management and the overall resilience and sustainability of supply chain activities. (Rajesh, 2020)

2.2 AI as Decision Support in Supply Chains

Artificial Intelligence has become a game-changing technology in supply chain management, fundamentally affecting organizational decision-making processes in complex and uncertain environments. The growing volume and complexity of supply chain data now exceed the capabilities of traditional rule-based systems and human judgment alone. (Toorajipour et al., 2021)

Historically, supply chain decisions have depended on human decision-making, statistical forecasting, and rule-based systems. While human judgment provides contextual expertise, it is vulnerable to biases and overconfidence, especially in volatile environments. (Perera et al., 2019) Rule-based systems utilize fixed logic but lack flexibility to dynamic markets characterized by shifting demands and disruptions. Statistical methods rely on historical patterns but often assume stable relationships that rarely stay the same. (Perera et al., 2019)

AI represents a qualitative advancement by learning directly from data, continuously updating models, and managing non-linear relationships across multiple variables simultaneously. (Toorajipour et al., 2021) As a result, AI is highly valuable for warehouses that require multi-supplier coordination, inventory optimization, and real-time responsiveness.

AI-driven decision support refers not to the complete automation of planning decisions, but to the use of AI-based systems that assist human decision-makers by transforming data into forecasts, recommendations, alerts, or suggested actions. This distinction is significant because supply chain planning decisions frequently involve uncertainty, incomplete information, and contextual aspects that algorithms may not entirely capture. Consequently, the value of AI decision support depends not only on technical capability but also on how users interpret, trust, and act upon AI-generated outputs. (Jarrahi, 2018)

Within warehouse replenishment, AI-driven decision support can assist planners by identifying demand patterns, estimating lead time variability, recommending order quantities, and flagging exceptions that require attention. Planners may choose to follow, adjust, or override these recommendations based on their experience, understanding of the operational context, and confidence in the underlying data. As a result, AI-supported replenishment constitutes a socio-technical decision process in which technology and human reasoning interact rather than operate independently. (Lee & See, 2004)

This perspective is central to the present study, as the research examines not only whether AI improves replenishment outcomes, but also how AI-supported recommendations are used in practice. In this context, AI may generate value through augmentation rather than automation by enhancing human planning capabilities while retaining final decision authority with planners. (Raisch & Krakowski, 2021)

2.3 AI Mechanisms: Predictive, Prescriptive, Real-Time Adaptation

AI-driven decision support within warehouses functions through three interconnected mechanisms: Predictive Intelligence, Prescriptive Intelligence, and Real-Time Adaptation.

Predictive Intelligence uses statistical algorithms and machine learning techniques to recognize patterns and trends from historical datasets, therefore supporting data-driven forecasts of future events and outcomes. (Božić, 2023) In the context of warehouses, this approach predicts demand trends, supplier lead times, and potential disruptions by analyzing sales history, market indicators, and operational data. (Lepenioti et al., 2020) Predictive models produce probabilistic forecasts that measure uncertainty, allowing planners to use forecasts as input to replenishment.

Prescriptive Intelligence goes beyond prediction by recommending specific actions. Using predictive model outputs, prescriptive systems apply mathematical programming, constraint-based optimization, and AI techniques to find optimal decisions across competing objectives. In warehouses, this includes recommendations for when and what to order, reorder quantities, safety stock levels, supplier selection, and replenishment timing, all of which balance service level targets, inventory carrying costs, working capital constraints, and warehouse capacity limitations. (Syed et al., 2025)

Real-Time Adaptation facilitates ongoing adjustment of decisions, replacing static, periodic reviews. These systems monitor live data streams from warehouse management systems, supplier networks, and market signals, detecting differences from expected conditions and automatically updating recommendations. (Lepenioti et al., 2020) In cases of sudden demand surges, unanticipated supplier delays, or abrupt market changes, these feedback mechanisms generate revised decision recommendations, notify planners of critical exceptions and adjust replenishment plans when conditions change.

2.4 Limitations of Traditional Approaches

Although human judgment, statistical forecasting, and rule-based systems have historically supported supply chain decision-making, each method indicates core limitations that become clearly visible in complex warehouse environments.

Human decision-making provides important contextual knowledge and expertise; however, it remains vulnerable to systematic biases, overconfidence, and misinterpretation of patterns, specifically in volatile environments with significant data variation. Studies indicate that planners frequently overestimate their forecasting precision and experience difficulties integrating multiple weak signals into logical predictions under uncertainty. (Perera et al., 2019)

Rule-based systems use fixed decision logic but lack the ability needed in dynamic markets characterized by shifting demands, short product life cycles, and frequent disruptions. (Perera et al., 2019) For example, a rule-based system may trigger a resupply order when inventory falls below safety stock; however, this approach does not take into consideration market signals, supplier constraints, and correlated demand patterns covering different product families.

Statistical methods advance beyond pure judgment by applying historical patterns, yet they rely on the assumption of stable relationships between variables and experience difficulties when these assumptions are invalidated by unforeseen events. (Perera et al., 2019) Structural breaks, such as pandemic-induced demand shifts or geopolitical supply disruptions, hinder these methods' adaptability, leading to systematic forecast errors and inaccurate replenishment decisions.

Collectively, these limitations generate a decision-making gap that AI-driven systems are well positioned to address, especially in warehouses that require simultaneous optimization involving multiple objectives and fast adaptation to progressing conditions.

2.5 Warehouses as Complex Decision Environments

Warehouses have been studied in the supply chain management literature as centralized coordination nodes designed to manage complexity arising from multi-supplier sourcing, demand uncertainty, and downstream distribution requirements. Early research conceptualizes warehouses as inventory consolidation points that enable coordination between upstream suppliers and downstream customers, with the objective of improving service performance while controlling inventory-related costs. (Shah & Goh, 2006) By centralizing inventory and decision-making, warehouses can exploit risk pooling effects, reduce overall safety stock requirements, and enhance responsiveness across the supply chain.

At the same time, the centralized nature of warehouses introduces substantial decision-making complexity. Warehouses are typically responsible for coordinating replenishment, allocation, and prioritization decisions across multiple products, suppliers, and downstream locations. These decisions must be made under conditions of demand volatility, uncertain lead times, capacity constraints, and competing performance objectives such as service levels, inventory holding costs, and working capital efficiency. As a result, warehouses operate as high-dimensional decision environments in which local decisions can have system-wide consequences. (Stadtler, 2005)

Several studies emphasize that traditional planning approaches struggle to cope with this level of complexity in centralized warehouse settings. Rule-based planning systems rely on predefined decision logic that lacks flexibility and responsiveness when demand patterns shift or disruptions

occur. Similarly, statistical forecasting methods typically assume stable relationships between historical demand patterns and future outcomes, an assumption that is frequently violated in dynamic supply chain environments. In warehouses, where multiple demand streams and correlated uncertainties intersect, these limitations become particularly pronounced and can lead to suboptimal inventory positioning and delayed responses to emerging risks (Christopher, 2016).

Human judgment has historically played a central role in managing warehouses, particularly in interpreting exceptions, coordinating across functions, and responding to unforeseen events. While experiential knowledge enables planners to incorporate contextual factors that are difficult to formalize, research shows that human decision-making is vulnerable to systematic biases, overconfidence, and cognitive limitations when dealing with large volumes of information and high uncertainty. In warehouse contexts, where planners must simultaneously evaluate multiple weak signals and trade-offs, these cognitive constraints can further reduce decision quality and consistency. (Perera et al., 2019)

Taken together, the literature characterizes warehouses as organizational settings where decision complexity, uncertainty, and interdependence are particularly concentrated. These characteristics make warehouses highly sensitive to the quality of decision support systems and planning processes used by organizations. Consequently, warehouses represent a relevant and promising context for examining the role of advanced decision support technologies, such as AI-driven systems, in augmenting human judgment, improving operational performance, and enhancing resilience and sustainability outcomes. This perspective provides a conceptual bridge to the literature on AI-driven decision support in supply chains, which is discussed in the following section.

2.6 AI-Supported Replenishment in Practice

AI-supported replenishment has become increasingly prominent in practical supply chain and warehouse planning solutions. (Toorajipour et al., 2021) No longer confined to experimental applications, AI is now widely promoted by supply chain software providers as a way to improve demand forecasting, inventory optimization, replenishment planning, and exception management. (Destino et al., 2022) These systems are meant to assist planners by generating more accurate

forecasts, recommending optimal order quantities, identifying inventory risks, and revising plans in response to changing demand or supply conditions.

In practical warehouse environments, AI-supported replenishment is typically dependent on integration with existing information systems such as warehouse management systems (WMS) and enterprise resource planning (ERP) systems. WMS platforms provide operational data on inventory movements, picking, replenishment assignments, and warehouse execution, while ERP systems often contain article, order, supplier, and financial data. As a result, the effectiveness of AI-driven decision support depends not only on the analytical model itself, but also on whether AI-generated recommendations can be connected to the systems where planning and execution data are maintained. (Mahroof, 2019; Jöhnk et al., 2021)

For instance, RELEX Solutions offers automated replenishment and forecasting tools that optimize stock levels, ensure high product availability, and enhance inventory turnover. The system incorporates automated parameter optimization based on historical sales data and forecast errors, while enabling exceptions to be managed through predefined rules or manual planner intervention (RELEX Solutions, n.d). Similarly, Blue Yonder emphasizes AI- and machine-learning-driven forecasting, probabilistic planning, exception-based workflows, and inventory balancing in its demand planning and replenishment solutions. Within this framework, replenishment planning is defined as the calculation of appropriate inventory quantities to achieve target service levels, balancing the risk of stock shortages against the costs of excess inventory. (Blue Yonder, n.d)

SAP provides AI-enabled planning capabilities through its SAP Integrated Business Planning and SAP Predictive Replenishment solutions. Predictive Replenishment is described as a machine-learning tool that forecasts demand, automates order proposals, and boosts service levels while limiting shortages and overstock. SAP Integrated Business Planning highlights the integration of real-time data, AI, and advanced planning tools to balance demand and supply and optimize inventory throughout the supply chain (SAP, n.d.). Collectively, these examples show that AI-supported replenishment is now embedded in commercial supply chain planning platforms, rather than remaining a theoretical or experimental concept.

However, although these solutions represent the practical availability of AI-supported replenishment capabilities, they do not fully address how planners utilize these systems in daily

warehouse decision-making. Specifically, there is limited understanding of how users interpret, trust, follow, or override AI-generated recommendations, and how factors such as master data quality, system integration, and planning routines influence effective utilization. This gap indicates the need to study AI-driven decision support not only as a technological capability, but also as a practical decision-making process within warehouse replenishment.

In addition to the availability of AI-supported replenishment tools, prior research indicates that effective AI utilization depends on both organizational and technical conditions. The AI readiness literature stresses that successful implementation needs alignment among data, technology, personnel, processes, and organizational structures, rather than only introducing an analytical tool (Jöhnk et al., 2021). This perspective is relevant for warehouse replenishment because AI-supported planning tools depend on data from operational and planning systems and must be integrated into established decision-making routines.

From a technical perspective, data quality and system integration are commonly identified as important foundations for AI-enabled decision support. AI systems require accurate, timely, and accessible data to produce reliable forecasts, recommendations, or alerts. In warehouse environments, relevant data includes inventory records, demand history, supplier information, lead-time assumptions, and warehouse execution data. (Raghuram & Singh, 2020) Furthermore, the smart warehousing literature highlights that digital warehouse technologies necessitate integration among information systems, operational processes, and human personnel to support effective decision-making (Mahroof, 2019; Jöhnk et al., 2021).

From an organizational perspective, effective AI utilization also depends on how users interpret, trust, and respond to system outputs. Research on trust in automation points out the necessity of appropriate reliance, in which users neither uncritically accept nor completely dismiss automated recommendations (Lee & See, 2004). The human-AI decision-making literature similarly indicates that AI can enhance human assessment by improving information processing and decision support, while human involvement continues to be essential for contextual interpretation and accountability (Jarrahi, 2018). This distinction between automation and augmentation highlights that AI may either replace specific tasks or enhance human decision-making capabilities (Raisch & Krakowski,

2021). Collectively, these perspectives offer a theoretical basis for analyzing the organizational and technical factors influencing AI effectiveness.

In addition to planning-oriented applications, the smart warehousing literature suggests that AI-enabled warehouse technologies are increasingly integrating decision support with operational implementation. Advanced warehouse systems support not only forecasting and replenishment recommendations, but also warehouse automation, picking optimization, routing, labor allocation, and coordination between digital systems and physical operations. (Mahroof, 2019) Although these applications go beyond replenishment decision-making, they are significant because they demonstrate how AI-supported planning is becoming integrated with broader warehouse execution and automation processes.

From a theoretical perspective, AI-supported replenishment is grounded in established principles of demand forecasting and inventory control. Classical inventory models, including reorder point policies, safety stock calculations, and newsvendor-type formulations, typically assume that demand distributions and lead times are either known or can be estimated with sufficient accuracy. These models also assume that order quantities and reorder points can be optimized by balancing the risk of stockouts against holding and ordering costs. Within this framework, demand forecasts and parameter settings, such as safety stock levels and service targets, serve as critical inputs to the replenishment decision.

AI-based planning tools modify these mechanisms by supplying improved forecasting abilities and adaptive parameter configuration. Machine learning-based demand forecasting leverages larger and more heterogeneous data sets, enabling forecasts to adapt more rapidly to changes in seasonality, promotional activities, and local demand patterns. Theoretical models suggest that these improvements should reduce forecast error and, as a result, decrease both stockouts and excess inventory. (Sen, 2026) Additionally, AI-enabled systems increasingly update safety stocks, reorder points, and exception rules dynamically, rather than relying on static parameters. This approach matches replenishment decisions more closely with the real-time, data-driven ideal envisioned in advanced inventory models.

The empirical component of this thesis therefore investigates both the availability of AI capabilities in commercial solutions and the ways in which warehouse planners perceive their

impact on forecasting quality, replenishment decisions, and operational outcomes. This analysis also considers whether the benefits predicted by theory are realized in the context of day-to-day replenishment planning.

3 Methodology

A qualitative case study research design is adopted to examine the implementation and use of AI-driven decision support in warehouses, as well as its influence on operational performance, risk resilience, and sustainability. This qualitative approach is suitable because the research aims to understand practices, perceptions, and contextual factors, rather than to measure the technical performance of specific algorithms. The case study design facilitates an in-depth analysis of AI usage within its organizational context, capturing the interactions among technology, processes, and human actors. (Barratt et al., 2011)

3.1 Research design

An interpretivist qualitative case study design is applied to examine AI-driven decision support in warehouse operations. This philosophical position is appropriate because the research attempts to understand how practitioners experience, interpret, and utilize AI in their decision-making, rather than testing predetermined causal relationships or quantitatively measuring technical performance.

Empirical material is collected through semi-structured interviews with multiple organizations that operate warehouses and use, evaluate, or develop AI-driven decision support tools. Instead of focusing on a single benchmark company, the study includes several organizations to develop a comparative understanding of AI support application throughout diverse warehouse contexts. The unit of analysis is the warehouse decision-making process and its associated planning activities. This encompasses AI-supported forecasting, replenishment, allocation, and related operational decisions, as well as the interpretation and implementation of these decisions by planners and other relevant stakeholders.

The purpose of this design is not formal benchmarking between companies, but rather to detect recurring patterns, differences, and practical conditions that influence the use of AI-driven decision support in warehouse settings. This approach enables an understanding of the phenomenon across organizational contexts while preserving the depth of each interviewee's experience.

A qualitative case study design is appropriate because AI-supported warehouse decision-making represents a complex and context-specific socio-technical phenomenon. (Mahroof, 2019) Its effective use depends on the relationship between data quality, planning routines, organizational

readiness, system integration, and human judgment, which can be properly understood only through in-depth qualitative inquiry. (Jöhnk et al., 2021)

Overall, the research design seeks to capture both the operational context and human interpretation of AI use in warehouses. This approach provides a strong foundation for analyzing how AI-driven decision support influences decision-making practices and operational outcomes in real organizational settings.

3.2 Data collection

Data collection primarily involved semi-structured interviews with stakeholders engaged in warehouse operations and planning. Stakeholders included supply planners, warehouse managers, and IT or data analytics specialists responsible for AI-based or advanced planning tools. Despite the diversity of firms, roles, sectors, and levels of AI maturity among the interviewees, the interviews continually focused on warehouse replenishment and related planning activities, including safety stock setting, reorder proposals, and slotting decisions. This approach ensured that the empirical material remained centered on replenishment-specific decision-making, even when respondents addressed broader AI initiatives within their organizations. The interview guide focused on four main themes: current planning and decision-making processes; the role and functioning of AI-driven decision support tools in warehouse operations; perceived effects on performance, resilience, and sustainability; and organizational and technical enablers and barriers to effective use. The full interview guide is provided in Appendices 8.1.1 and 8.1.2 and was used to structure all interviews while still allowing for follow-up questions designed to each interviewee's role and context.

The semi-structured interview format was selected in order to ensure a consistent thematic structure while allowing respondents flexibility to elaborate on their experiences, interpretations, and real examples (Britten, 1995). This approach was particularly important due to the exploratory nature of the study and the variation in roles, organizational contexts, and levels of AI maturity among participating organizations.

Empirical material was collected from multiple organizations operating in warehouse and logistics contexts, rather than from a single focal company. This approach allowed the capture of a wider

range of experiences and facilitated comparison of how AI-driven decision support is understood, implemented, and used across several settings. The inclusion of multiple organizations aimed to strengthen the empirical basis and detect recurring patterns and meaningful differences between respondents, rather than to conduct a formal benchmark analysis.

This thesis evaluates operational performance using qualitative indicators instead of direct access to internal key performance indicator (KPI) reports. The interview questions specifically addressed perceived changes in service levels, stockouts, excess inventory, lead time predictability, workload distribution, and planning efficiency after the implementation of AI-supported tools; so respondents often referenced specific performance metrics within their organizations. However, these metrics were typically reported in aggregate or narrative form rather than systematically collected as numerical data.

Consequently, the study documents how practitioners describe and evaluate operational performance in AI-enabled warehouses and how they associate these outcomes with AI-driven decision support. However, it does not statistically validate performance changes or differentiate them from other simultaneous operational initiatives.

A total of 9 semi-structured online interviews were conducted between March and May 2026, as well as 4 in-person discussions. Respondents were selected using purposive criterion-based sampling. Inclusion criteria specified that participants must either operate a warehouse or work in a supply chain planning role, and their organization must be actively using or evaluating AI-driven decision support tools. No restrictions were imposed regarding industry sector, organizational size, or AI maturity level. Recruitment involved contacting as many qualifying organizations as possible within the available timeframe. Two respondent categories were included: users and solution providers involved in designing, implementing, or advising on such systems; which some preferred to remain anonymous. This dual perspective facilitated analysis of both the intended functionality of AI tools and their practical interpretation in real operational settings. Table 1 below provides a summary of the interview respondents:

#	Respondents	Company	Sector	Role	Type	Format	Date	Interview Duration
1	Anders Sjögren	AstraZeneca	Pharma	Associate Director Global Supply Chain Process	User	Online	March 2026	40 mins
2	Mikael Engberg	Postnord	Logistic	Previous Manager of Logistics Development	User	Online	May 2026	45 mins
3	Claudio Muñoz Percheron	L'Oréal	Consumer Goods	Core Supply Chain Business Expert	User	Online	May 2026	35 mins
4	Karim Nouria	SICS AI	Technology, IT	Founder	Solution Provider	Online	May 2026	25 mins
5	Bernt Hylen	Consafe	Logistic, IT	Product Owner AI and Digital Twin	Solution Provider	Online	May 2026	35 mins
6	Anonymous Interviewee 1	Anonymous Company 1	Construction	Warehouse Manager	User	In-Person	Feb 2026	45 mins
7	Julia Lindblom	Göteborgs Posten	Journalism	Investigative Journalist	N/A	In-Person	April 2026	1 hour
8	Anonymous Interviewee 2	Anonymous Company 2	FMCG	Regional Distribution	User	Online	March 2026	40 mins

				Center Manager				
9	Mikael Engberg	SWECO	Consultancy	Senior Logistics Consultant	User	Online	May 2026	45 mins
10	Anonymous Interviewee 3	H&M	Retail	Warehouse Responsible	User	In-Person	April 2026	2 hours
11	Filip Höcker	Consafe	Logistic, IT	Sales Executive	Solution Provider	In-Person	April 2026	1 hour
12	Anonymous Interviewee 4	Unilever - Dubai	FMCG	Logistics Lead Responsible	User	Online	May 2026	40 mins
13	Joakim Waern	Waernhouse Automation	Logistic	Business Owner	User	Online	April 2026	1 hour

Table 1: Table showing the list of interviewees who participated in this study

In addition to interviews, participation in the Plan Konferensen 2026 conference in Gothenburg on April 21 and 22 served as an important additional data collection method. The event brought together speakers and companies active in logistics, planning, and warehousing. As artificial intelligence was a central topic, conference participation provided exposure to current industry discussions and practices related to AI implementation in warehouse settings. Conference participation provided additional contextual and observational findings that supported the interview material. It enabled direct interaction with practitioners, facilitated access to additional respondents, and situated the interview findings within wider industry developments. These insights served as a complementary source to enrich understanding of the empirical context, rather than as a primary source of evidence.

The interview guide addressed both users of AI-driven decision support tools and solution providers involved in designing or implementing such systems. This approach enabled the capture of perspectives from both the user and design sides of AI-supported warehouse decision-making.

Including these respondent categories broadened the analysis by revealing not only intended functions of AI tools, but also their interpretation and use in practice.

All interviews were carried out to generate rich qualitative material suitable for inductive coding and interpretive analysis. Particular attention was given to collecting concrete examples, descriptions of decision situations, perceived benefits and challenges, and reflections on the interaction between AI recommendations and human judgment.

3.3 Data analysis approach

Interview data were transcribed and analyzed using an iterative qualitative coding process. Instead of a purely descriptive thematic analysis, the study adopted an interpretive and inductive approach based on the Gioia methodology. This method is well-suited for identifying patterns in interview material while maintaining a clear connection between participants' expressions and researchers' higher-level interpretations (Gioia et al., 2013).

The analysis started with a close reading of the interview transcripts to gain familiarity with the material and identify meaningful statements regarding the use of AI-driven decision support in warehouse contexts. During this stage, initial codes were assigned to excerpts that reflected respondents' terminology, experiences, and descriptions of the phenomenon. These first-order codes remained closely aligned with the informants' language to preserve empirical richness and avoid premature abstraction (Gioia et al., 2013).

Subsequently, first-order codes were compared across interviews and grouped into broader second-order themes. At this stage, the analysis shifted from informant-centered descriptions to interpretive categories that captured recurring patterns and explanatory concepts in the data. (Magnani et al., 2023) Examples of these themes included AI as a supporting mechanism, the importance of lead time, the role of master data, and the continuing need for human judgment.

Where appropriate, related second-order themes were synthesized into more general aggregate dimensions to structure the findings and clarify the broader analytical contribution of the study. This step facilitated the transition from raw interview material to a coherent data structure, showing how operational experiences with AI-driven decision support related to broader organizational and decision-making implications. (Jimenez et al., 2024)

Concurrently, the analysis was guided by the study's conceptual background and research questions. Concepts such as predictive, prescriptive, and real-time adaptive AI, along with operational performance, resilience, sustainability, and organizational and technical enabling factors, served as analytical lenses during interpretation. These concepts were not imposed as fixed categories but functioned as sensitizing concepts, supporting interpretation while allowing new themes to emerge inductively from the empirical material. (Gioia et al., 2013)

Interviewee	Company	Position	Concept Number	First-Order Concepts	Evidence	Interpretation
Anders Sjögren	Astrazeneca	Associate Director Global Supply Chain Process	1	AI is used indirectly, not as a core decision-maker	"We do use AI indirectly, tools like Copilot or our internal ChatGPT, but they mainly support daily tasks rather than core supply chain decisions"	AI currently supports general productivity rather than driving replenishment decisions directly
			2	Replenishment lead time is the key use case	"We also have another use case with Era that is further along This relates to replenishment lead time"	The interviewee identifies replenishment lead time as their most advanced AI application
			3	Lead time is a central planning variable	"replenishment lead time, the time from when we start planning a product to when it is available in the market"	Lead time spans entire process from planning to stock availability, core to replenishment
			4	Transportation lead time matters	"Part of that lead time is transportation For example, if we produce in Sweden and ship to Taiwan by boat, it might take around six weeks"	Transportation forms significant portion of total replenishment lead time
			5	Long transport routes create uncertainty	"if we produce in Sweden and ship to Taiwan by boat, it might take around six weeks"	International routes introduce significant variability in planning assumptions
			6	AI could help verify lead time assumptions	"We've introduced an AI tool to validate whether our planned lead times match reality"	AI specifically targets gap between designed/planned vs actual lead times
			7	Master data is crucial	"a key factor is having high-quality master data If your data isn't reliable, applying AI won't help; it will just produce unreliable outputs"	Data quality determines AI effectiveness—bad data produces unreliable outputs
			8	AI may help reduce waste	"we can sometimes redirect products to markets with lower requirements However, if a product can't be sold, it must be destroyed"	Better lead time accuracy could reduce unsellable inventory nearing expiry
			9	AI is not replacing human decision-making	"they mainly support daily tasks rather than core supply chain decisions So far, there hasn't been a direct impact"	Human oversight remains essential for supply chain decision authority
			10	AI is a supporting tool, not a full solution	"AI is not a magic solution, it won't fix everything"	AI augments rather than automates complex supply chain processes
			11	Planning is complex and parameter-heavy	"There are many parameters involved... balance the risk of unsold products with production efficiency"	Replenishment involves multiple interdependent variables requiring human judgment

Table 2: Table representing one interview's coding into First-Order Concepts

The coding process was iterative rather than linear. As the analysis progressed, codes and themes were continuously reviewed, merged, refined, or separated in response to new insights. Similarities and differences across respondents were compared to identify both converging opinions and meaningful variations between organizations, roles, and experiences. This iterative comparison

enhanced the analytical precision of the study by assuring transparency and systematic development from raw data to final themes. (Roberts et al., 2019)

	First-Order Concepts	Second-Order Themes	Aggregate Dimensions
1	AI is used indirectly, not as a core decision-maker	Human Decision-Making Support	Human-AI Coordination
2	Replenishment lead time is the key use case	Lead Time Uncertainty Management	AI-Supported Planning and Replenishment
3	Lead time is a central planning variable	Lead Time Uncertainty Management	AI-Supported Planning and Replenishment
4	Transportation lead time matters	Lead Time Uncertainty Management	AI-Supported Planning and Replenishment
5	Long transport routes create uncertainty	Lead Time Uncertainty Management	AI-Supported Planning and Replenishment
6	AI could help verify lead time assumptions	Lead Time Uncertainty Management	AI-Supported Planning and Replenishment
7	Master data is crucial	Data Reliance	AI-Supported Planning and Replenishment
8	AI may help reduce waste	Measurable Replenishment Benefits	Operational Value Creation
9	AI is not replacing human decision-making	Human Decision-Making Support	Human-AI Coordination
10	AI is a supporting tool, not a full solution	Human Decision-Making Support	Human-AI Coordination

Table 3: Table representing the iteration leading to Second-Order Themes and Aggregate Dimensions

The findings were organized around the most prominent themes that emerged from this process. The coding table (full version in the appendix 8.2) serves as an audit trail, illustrating how selected empirical statements were translated into first-order codes, interpretive themes, and broader analytical insights. This approach balances empirical closeness with conceptual development, consistent with an interpretive qualitative research design.

Aggregate Dimensions	Second-Order Themes	First-Order Concepts
Automation, Scaling and Societal Implications	AI Ethical & Labor Issues	5
	Robotic Picking Automation	13
	Scalable AI Expansion	5
	Workforce Transformation Concerns	3
	Warehouse Optimization	19
Automation, Scaling and Societal Implications Total		**45**
AI-Supported Planning and Replenishment	End-to-End Supply Planning	5
	Lead Time Uncertainty Management	6
	Predictive Slotting & Layout	6
	Data Reliance	3
	Generating Replenishment Proposals	3
	Productivity & Seasonality Improvements	4
	Automated SKU Replenishment	8
AI-Supported Planning and Replenishment Total		**35**
Adoption and Implementation Conditions	Emerging Adoption Pressures	7
	Implementation & Change Barriers	5
	System Integration Responsibilities	3
	Organizational Adoption Conditions	7
	Cross-Functional Coordination	2
	Organization-Wide AI Adoption	3
	Adoption & Prediction Challenges	5
Adoption and Implementation Conditions Total		**32**
Operational Value Creation	Business Case Justification	5
	Space Constraint Optimization	2
	Measurable Replenishment Benefits	13
	Improving Operational Performance	6
Operational Value Creation Total		**26**
Human-AI Coordination	Predictive Exception Alerting	4
	Trust & Data Quality Challenges	3
	Human Decision-Making Support	6
Human-AI Coordination Total		**13**
Total Number of First-Order Concepts		**151**

Table 4: Table summarizing the First-Order Concepts, Second-Order Themes and Aggregate Dimensions

3.4 Research quality and limitations

Credibility is enhanced through triangulation of interviews and other sources, and, where appropriate, by conducting member checks in which summaries of key findings are shared with selected respondents for feedback. (Carter, 2014) However, the study has several limitations. The focus on a single or small number of cases restricts the potential for statistical generalization, and the analysis is contingent on the level of access granted by participating organizations and the willingness of practitioners to share their experiences.

The emphasis on a limited number of cases and the use of a qualitative, interpretive design limits the potential for statistical generalization to the wider population of warehouses implementing AI. Instead, this study aims for analytical generalization by connecting perceived patterns to established theories on decision support systems, socio-technical change, and warehouse management, and by developing propositions that may be applicable to other contexts with comparable operational characteristics and degrees of AI maturity. In addition to sample size and research design, several contextual factors influence the generalizability of the findings. The organizations interviewed are primarily based in Europe and operate in sectors including fast-

moving consumer goods, pharmaceuticals, fashion, and logistics services. These organizations manage warehouses that either utilize or are considering the adoption of AI-supported planning tools. Therefore, the results are most applicable to medium- and large-scale warehouses with similar levels of digitalization, data availability, and process formalization. The findings may be less relevant to very small, low-automation warehouses or settings with limited IT infrastructure.

Thus, the findings should not be generalized to all warehouses but should be considered relevant primarily for those with comparable operational complexity, replenishment responsibilities, and readiness to implement AI-driven decision support.

Consequently, practitioners and researchers should interpret the findings as theoretically informed insights and propositions intended to guide subsequent empirical investigation, rather than as representative estimates of average effects within the logistics sector.

The internal validity of this study is constrained by its qualitative, interview-based case design. Instead of directly measuring the causal effects of artificial intelligence (AI) regarding operational performance, the analysis depends on respondents' descriptions of changes over time and their interpretations of how AI-supported tools affect service levels, inventory, and workload. As a result, the study identifies perceived mechanisms and patterns that indicate a potential relationship between AI-driven decision support and performance outcomes. However, it does not empirically isolate or quantify the net causal impact of AI in comparison to other simultaneous changes, such as process redesign, demand fluctuations, or organizational initiatives.

This thesis conceptualizes AI as a contributory factor within broader socio-technical transformations, rather than as an isolated, independently verifiable cause of performance improvement. Accordingly, assertions regarding "AI improving performance" are analytically inferred from convergent interview evidence and should not be interpreted as statistically validated causal estimates.

Finally, there is an important distinction between perceived and actual operational improvements. In this study, the reported effects of AI on service levels, inventory, lead time predictability, and workload are derived from managers' and planners' descriptions and evaluations of operational changes, rather than from independently collected key performance indicator (KPI) data.

Consequently, the findings reflect perceived operational improvements in organizations implementing AI-supported tools, but do not confirm the magnitude or existence of these improvements through audited performance measurements.

3.5 Ethical Considerations

Ethical considerations guided all stages of the research process. Interview participants received information regarding the study's purpose and provided their consent prior to the interviews. Respondents were offered the option to remain anonymous, and company-sensitive information was managed with strict confidentiality. Interview data and transcripts were securely stored and used exclusively for academic purposes. The study's objective is to examine broader experiences and practices related to AI-supported warehouse planning, rather than to assess individual performance or reveal sensitive operational details.

4 Findings

The findings are organized according to five aggregate dimensions identified through the inductive coding process using the Gioia methodology (see Appendix 8.3). Each section introduces the aggregate dimensions and their underlying concepts, supported by direct quotations from interviewees. The analysis progresses from the positioning of AI in relation to human decision-making, to its role in supporting planning, the operational value generated, the barriers and enablers encountered during implementation, and concludes with the emerging and future capabilities under development and deployment.

4.1 Human-AI Coordination

A consistent pattern identified across all interviews was the characterization of AI as a supporting mechanism rather than as an autonomous or decisive actor in warehouse replenishment. This theme was present irrespective of company size, sector, or the maturity of AI adoption, and represented the most widely shared perspective among interviewees. In other words, the distinction between AI functions and areas where humans retain authority was repeatedly emphasized, with interviewees elaborating in detail on the conditions which determine this boundary.

4.1.1 AI as a Decision Support Tool Rather Than a Decision-Maker

Across the interviews, respondents described how AI tools are positioned within their organizations relative to human decision-making. Anders Sjögren of AstraZeneca explained that AI use in his organization is primarily indirect: “We do use AI indirectly, tools like Copilot or our internal ChatGPT, but they mainly support daily tasks rather than core supply chain decisions,” adding that “so far, there hasn’t been a direct impact” on replenishment decisions. He further noted, “AI is not a magic solution, it won’t fix everything,” reflecting a deliberately measured organizational stance toward adoption. From a solution provider perspective, Mikael Engberg of Consafe emphasized that human validation of AI outputs remains essential: “You need to look at whether it is actually reasonable.” At L’Oréal, this principle is formalized as an organizational requirement, with Claudio Muñoz Percheron stating, “There always needs to be a human in the middle checking the information coming from AI.” This view was not limited to organizations already using AI. An anonymous respondent from Unilever UAE, whose warehouse had not yet adopted AI, described the anticipated role of such tools as one of data consolidation and visibility

support: “Having AI bring that data together and provide visibility would be really valuable, not just for operational decisions but also for cost control, spotting inefficiencies across all aspects of the operation faster and more clearly.”

These accounts, drawn from organizations at different stages of AI adoption and across both user and provider roles, collectively suggest that AI is consistently framed as a decision-support mechanism rather than an autonomous decision-maker. The boundary between AI-generated recommendations and human judgment appears to be both practically maintained and, in some cases, formally institutionalized within organizational workflows.

4.1.2 The Shift from Routine Work to Exception Management

Several interviewees described a transition from manual, repetitive calculations to supervisory and exception-based work. One interviewee, who asked to stay anonymous, explained: “Instead of planning manually, I supervise and validate what the system suggests,” and further described their day-to-day responsibilities as follows: “My role is to review exceptions like sudden demand spikes, anomalies, or supplier constraints, and adjust when needed.” This change was also described from an operations point-of-view: “The difference is that we now focus more on managing exceptions rather than processing routine decisions.” The redistribution of workload was clearly acknowledged as a qualitative, rather than quantitative, change: “In terms of workload, it’s shifted rather than reduced. I spend less time on manual calculations and more time reviewing exceptions and coordinating with suppliers.”

Although AI does not replace human decision authority, these accounts indicate that it fundamentally reshapes how planners allocate their attention and time. The human role in AI-supported replenishment is not diminished but redefined, shifting from data processor to decision supervisor in most cases.

4.1.3 Complexity and the Continued Indispensability of Human Judgment

Several interviewees stated that the complexity of replenishment planning makes full automation both difficult and undesirable. Planning calls for navigating numerous interdependent variables that are difficult to fully encode into algorithmic logic. As Sjögren expressed, “There are many parameters involved […] balance the risk of unsold products with production efficiency,” drawing attention to the need for continuous trade-off decisions based on the situation and company

priorities. Another interviewee emphasized, “A big part of the job is balancing availability with cost. We can’t afford stockouts because that delays projects, but overstocking ties up capital and space.” Even during extreme disruptions, human intervention remained essential: “However, in extreme situations like sudden supply chain disruptions, we still rely on manual intervention and experience.” The experience of the Unilever interviewee further supports this point. Despite managing major supply chain disruptions such as port congestion, border delays, and rising costs without AI support, the team maintained operational continuity and achieved strong business results: “We managed to ensure the safety of our people while continuing to deliver market needs in a stable manner… despite all of that, we delivered well and actually achieved good growth in Q1.” Interestingly, a H&M site visit allowed us to see an organizational approach that clearly favors human expertise over AI, stating that their planning and warehouse operations rely on “really smart persons,” thus positioning human expertise as an organizational strategy rather than a temporary measure.

These accounts portray tensions between service levels and holding costs, and between risk and efficiency, as areas where human judgment remains needed and difficult to fully codify into algorithms. Together, they suggest that experienced practitioners continue to play a central role in maintaining operational performance under pressure, even as AI adoption increases.

4.1.4 Uneven Trust and the Need for Explainability

Interviewees described trust in AI-generated recommendations as variable both within and across organizations. One interviewee observed, “Some planners initially relied too much on the system, while others didn’t trust it enough,” highlighting that adjusting appropriate reliance remains an ongoing organizational challenge. A primary factor playing a role in this uneven trust was identified as the lack of explainability in AI outputs: “In terms of improvements, better explainability would help, understanding why the system makes certain recommendations would make it easier to trust and use effectively.”

The effectiveness of human-AI coordination is therefore influenced by the degree to which planners trust AI-generated recommendations. This observation shows that even if a system is technically capable of giving accurate results, users may still not trust the output without

understanding how the system arrived at it, pointing to explainability as a precondition for effective utilization, not merely a desirable feature.

4.2 AI-Supported Planning and Replenishment

The second aggregate dimension studies the integration of AI within the planning and replenishment process. This includes lead time management, data integration, replenishment proposal generation, warehouse slotting, and the significance of data quality. Interviewees consistently indicated that AI enables a structured, data-driven, and preventive approach to replenishment which would be unfeasible, in most cases, to manually achieve given the scale and complexity of operations.

4.2.1 Lead time uncertainty as a primary AI application

Lead time management represents one of the most concrete and frequently cited applications of AI in warehouse replenishment. Sjögren of AstraZeneca identified replenishment lead time as their most advanced AI use case: “We also have another use case with Era that is further along. This relates to replenishment lead time, the time from when we start planning a product to when it is available in the market.” In this context, lead time involves the entire period starting with initial planning to market availability, with transportation identified as a major source of variability: “Part of that lead time is transportation. For example, if we produce in Sweden and ship to Taiwan by boat, it might take around six weeks.” The inconsistency between planned and actual lead times was highlighted as an important issue, prompting the deployment of AI tools to validate and align these timelines: “We’ve introduced an AI tool to validate whether our planned lead times match reality.” Another interviewee further confirmed that AI has improved lead time predictability: “Lead times are more predictable because we plan earlier and more accurately.”

This use case illustrates a wider recognition that planning system assumptions may differ from actual operational realities, and that AI can serve a verification function to identify and correct such discrepancies, improving not just forecast accuracy but the reliability of the planning baseline itself.

4.2.2 Generating replenishment proposals by combining various data

An important capability described by interviewees was the ability of AI systems to synthesize several data streams into structured, actionable replenishment proposals. One of the anonymous interviewees described a system that "integrates historical consumption data, project timelines, and supplier lead times" to produce daily outputs: "It generates replenishment suggestions, basically recommended order quantities and timing for each SKU." He also described a more advanced version at scale, supported by "an AI-based forecasting and replenishment system that processes sales data from retailers, promotional plans, seasonality, and external factors like holidays or even weather patterns." This system went further in its operational integration: "The system generates daily replenishment proposals for each SKU, which are automatically integrated into our ordering system." Beyond proposals, the system also performed predictive safety stock management: "It also dynamically updates safety stock levels based on demand variability and supplier reliability," allowing adaptive inventory control rather than fixed parameter-setting. The level of detail these proposals have was also remarkable: "The system predicts demand per SKU, per location, and even adjusts for promotions or campaigns, which are very common in FMCG." As the same interviewee noted, this level of resolution would be operationally impossible without AI: "That's something we couldn't realistically manage manually across thousands of SKUs."

Taken together, these accounts point to a meaningful shift in how replenishment proposals are generated, from planners manually consolidating data across systems to AI synthesizing multiple streams into granular, automated outputs. Interviewees emphasized that the scale and resolution of these proposals, down to individual SKUs, locations, and promotional events, would be difficult to achieve reliably through manual planning alone.

4.2.3 Exception flagging as a planning support mechanism

In addition to generating proposals, AI systems actively flag anomalies and exceptions to direct planners' attention to critical issues. An interviewee who wished to remain anonymous explained: "The system also flags exceptions, like unusual demand spikes or potential shortages, which helps me focus my attention where it's actually needed." Similarly at L'Oréal, exception detection is integrated into the order waving process: "The AI will analyze what kind of problems would happen if you wave these orders," identifying operational capability issues such as "if there is one SKU that does not have a picking location." Exception alerts also address shipment timing, with

the system highlighting risks such as “the truck is late,” which enables early intervention before cutoff risks occur. Real-time data is essential to this responsiveness: “The system continuously updates forecasts based on real-time sales data, so we can react quickly to changes in demand.”

Collectively, these features position AI as a tool for continuous monitoring of plan integrity and for surfacing deviations for human review, rather than for generating complete autonomous plans. The value of exception flagging lies not in replacing planner judgment but in ensuring that attention is directed where it is most needed, compressing the time between a disruption occurring and a planner becoming aware of it.

4.2.4 Predictive slotting and warehouse layout optimization

AI also assists with warehouse layout and product placement optimization within replenishment processes. Mikael Engberg, a previous PostNord Manager of Logistics Development, described a solution focused on “the buffer zone and the picking zone,” where AI improved product placement and streamlined replenishment flows. The main mechanism was ABC classification: “What it really did was put items into three categories: A, B, and C,” enabling the warehouse to focus on location quality for fast-moving items while managing slower products appropriately. Historical and forecast data drove these classifications: “We had a solution that looked at historical data” to decide how articles should be positioned, together with predictions extended to new SKUs by proxy: “If there was a new article, we could look at a similar product from before.” Bernt Hylén, Product Owner AI and Digital Twin at Consafe, described a similar logic built into the Astro WMS, where “warehouse mapping optimizes storage by frequency and associations while respecting constraints like heavy items first to avoid damage,” and where AI includes “article associations for optimal placement,” a logic comparable to recommender system logic applied to picking efficiency. Route optimization was also described as an operational feature: “Route optimization solves the traveling problem for shortest pick paths, but requires distance mapping,” alongside order batching, described as “combining orders to fill pallets efficiently.” The primary motivation behind the PostNord implementation was clearly stated: “It also helped keep the distance short when picking items. That was the main reason for us.”

Across these accounts, the primary objective was consistent: maximizing picking productivity by reducing travel distance and improving product positioning. In the cases described here, AI-

supported slotting and layout optimization extend data-driven classification and prediction from replenishment timing and quantities to the physical organization of the warehouse itself.

4.2.5 Data quality as a foundational condition

Data quality is fundamental to all AI planning capabilities, serving as a prerequisite for reliable outputs. The AstraZeneca interviewee emphasized: “A key factor is having high-quality master data. If your data isn’t reliable, applying AI won’t help; it will just produce unreliable outputs.” Unilever’s employee identified data quality as an ongoing operational constraint: “The quality of that data is itself a challenge […] Manually gathering and interpreting data to make decisions is time-consuming and creates inefficiencies for the team.” Unlike others who discussed past data issues, this example highlights a current data readiness gap that organizations must address before adopting AI. Similarly, another anonymous interviewee described forecast accuracy as highly dependent on data quality and integration: “Forecast accuracy depends heavily on data quality and integration.” Planning was also characterized as complex and parameter-intensive, with “many parameters involved” that require attentive management and maintenance.

These perspectives are consistent with the understanding that AI systems amplify the quality of their inputs rather than compensate for deficiencies. Data quality is therefore not only a technical requirement but a foundational precondition, one that must be actively managed before and during AI adoption, rather than assumed to be in place. Interviewees framed this as significant for organizations still in early stages of AI readiness, where data infrastructure gaps may represent a more immediate barrier than access to AI tools themselves.

4.2.6 Replenishment logic

From the perspective of solution providers, Consafe’s Bernt Hylén identified three distinct replenishment logics within the Astro WMS: “three main types: order-driven, low-watermark, and predictive replenishment.” First, order-driven replenishment “triggers assignments when you release picks for known orders,” serving as a reactive mechanism closely aligned with confirmed demand. Second, low-watermark replenishment “creates assignments when inventory falls below a set level in pick locations,” acting as a safety net to maintain pick-face availability in high-velocity settings. Lastly, predictive replenishment uses a different approach, exemplified with a real case: “It forecasts demand using historical patterns, so warehouses replenish in the morning

when staff is available and focuses on picking later," and is especially effective when "orders arrive late in the afternoon but must be ready early next morning."

These varied replenishment logics illustrate that AI-supported planning is not applied uniformly but is instead configured to match the operational rhythm and demand profile of each warehouse context. The coexistence of reactive, threshold-based, and predictive logics within a single WMS suggests that effective AI-supported replenishment requires not just technical capability but deliberate configuration choices aligned with operational realities.

4.3 Operational Value Creation

The third aggregate dimension involves outcomes attributed by interviewees to AI-driven decision support, including measurable improvements in service levels, inventory efficiency, workload quality, transport performance, and reduction of waste. These outcomes were described qualitatively through interviewees' reflections on operational change, as well as quantitatively through references to verifiable metrics and before-and-after comparisons.

4.3.1 Service level improvements and inventory optimization

Improvement in service levels emerged as the most consistently reported operational benefit. One of the anonymous interviewees observed "a noticeable improvement in service levels, fewer urgent shortages at construction sites," crediting this to better demand anticipation rather than reactive ordering. Similarly, another interviewee mentioned significant improvements at a larger scale: "Service levels have improved significantly, especially during promotions where demand is harder to predict," highlighting that AI's value is most apparent during periods of high demand volatility. Bernt Hylén of Consafe verified these outcomes across the WMS product's user base: "These methods improve service levels, inventory control, lead times, and workload differently based on business needs." Regarding inventory, AI-supported replenishment contributed to reductions in excess stock: "We've reduced excess inventory, particularly for slow-moving or seasonal items," with safety stock levels adjusted dynamically rather than set statically. Automation also enabled greater scale: "A large share of replenishment orders is generated automatically, which allows us to handle a much higher volume without increasing staff."

Taken together, these accounts point to two distinct but related value mechanisms: AI improves planning precision during volatile periods, reducing reactive shortages and excess stock, while simultaneously enabling greater operational scale without proportional increases in headcount. In the cases described, the combination of improved service levels and inventory optimization suggests that AI-supported replenishment generates value not only through better decisions but also by expanding the capacity to make such decisions consistently across larger product ranges.

4.3.2 Waste reduction and sustainability outcomes

Several interviewees linked AI-supported replenishment to concrete reductions in waste, particularly for products with short shelf lives or at risk of expiry. Anders Sjögren of AstraZeneca described instances in which improved lead time accuracy prevented situations in which "if a product can't be sold, it must be destroyed," noting that while products nearing expiry could sometimes be redirected to markets with lower regulatory requirements, complete avoidance of such situations is preferred. Another interviewee confirmed this outcome: "There's been a clear impact on waste reduction, especially for perishable goods," and added that "the system helps us avoid overstocking products that risk expiring." Transport consolidation was also identified as a sustainability-related benefit: "We're also consolidating shipments more effectively, which reduces the number of deliveries overall," and "We've also improved transport efficiency by consolidating shipments and reducing urgent replenishment orders." According to Unilever's employee, waste reduction and cost control were identified as main reasons to consider AI adoption, noting that route planning and optimization could "significantly increase our effectiveness and efficiency in deliveries while reducing cost and waste." This perspective shows that waste reduction is a recognized operational gap motivating AI adoption among those still using manual processes.

Karim Nouri, SICS AI's founder, further associated their robotic picking system with sustainability: "As long as you provide high efficiency, you use resources and energy more efficiently, so of course it contributes to sustainability."

However, an interesting and contrasting perspective was raised by Joakim Waern, who warned against the tendency to selectively emphasize the benefits of emerging technologies while overlooking their costs. Waern challenged the notion that AI's contribution to sustainability is

purely positive, drawing an analogy to electric vehicles to argue that widespread adoption of new technologies does not necessarily result in environmental progress; rather, it may simply redistribute consumption instead of reducing it: “You are not a hero for buying an electric car; same thing for AI.” In the context of AI, this reasoning implies that organizations highlighting efficiency gains from AI-driven operations may also be generating substantial energy and computational overhead that remains unacknowledged. Waern emphasizes this tension by contrasting concern over single-use plastics with the simultaneous operation of thousands of AI-driven processes: “If you’re having concerns about using straws in your drink while running a thousand animations, then you are not doing your part.” He further observes that widespread demand can suppress close inspection of negative effects: “everyone wants it, no one will stop it.”

Collectively, these accounts portray sustainability as an indirect benefit of improved process efficiency, rather than as a primary design objective of AI implementations. At the same time, Waern’s critical perspective introduces an important counterpoint, suggesting that characterizing AI as inherently green or efficient may reflect trend-driven optimism rather than a thorough environmental assessment. This tension between reported efficiency gains and unacknowledged computational costs emerges in the material as a relatively novel and underexplored theme.

4.3.3 Order quantity optimization and improvements

In addition to inventory management, interviewees indicated that AI increases the accuracy of individual order decisions. One of the anonymous interviewees stated that “it also optimizes order quantities based on supplier constraints and transport efficiency,” therefore aligning ordering logic with logistical realities. Delivery visibility was identified as a secondary benefit: “Internally we respond faster because we have better visibility,” which allows faster internal coordination even when external lead times remain unchanged. For AI robotic systems, SICS AI’s founder emphasized that “you get excellent transparency” as a key operational feature, and that “with this increased supply chain transparency, better forecasting and prediction ability” became possible as a result of operational data generated by the system. Financial benefits were also confirmed by the warehouse manager whose organization implemented the slotting AI: “When we looked at the time spent before and after implementation, we earned money from it.”

Taken together, these accounts suggest that AI-driven order optimization generates value beyond inventory levels alone, extending to logistical alignment, internal responsiveness, and measurable financial returns. Notably, the financial benefit cited here was arrived at through a direct before-and-after comparison in one case, lending additional empirical weight to efficiency claims that are often reported only qualitatively in the literature.

4.3.4 Workload redistribution and workforce implications

Interviewees described the impact of AI on planners' and warehouse workers' daily activities in nuanced terms. Unilever's employee described manually gathering and interpreting data as time-consuming and inefficient for the team: "Manually gathering and interpreting data to make decisions is time-consuming and creates inefficiencies for the team." Recently appointed to his role, he uses basic tools to consolidate data and support decision-making: "I'm currently using some basic tools to bring data together and create some visibility for decision-making." This example illustrates the starting point from which AI-driven workload redistribution would indicate a meaningful improvement, providing a clear operational contrast to the efficiency gains reported by other interviewees.

One such example is the Consafe interviewee who explained that predictive replenishment spreads workload across shifts: "This spreads workload, cuts overtime costs, and improves work-life balance," therefore improving both productivity and working conditions. At L'Oréal, AI-supported workforce distribution was described as an active planning tool: "The AI can propose a distribution of your workforce in the different workshops," providing real-time labor allocation support.

Beyond workload optimization, interviewees also discussed the broader labor implications of warehouse enhancements. The Consafe interviewee stated that "automation, not planning tools, poses the biggest job threat by replacing manual labor," while SICS AI's founder confirmed that their system is "primarily replacing manual pickers in the warehouse." Productivity gains were quantified: "We pick about four times faster than a human, depending on the objects," and the system can "work 24/7," making labor substitution an important consideration where robot deployment is feasible. Cross-functional collaboration was also identified as an operational

outcome: “Collaboration has also increased, we work more closely with sales and marketing teams because their input, like promotions, directly affects the AI forecasts.”

These accounts reveal two distinct labor dynamics operating in parallel. Planning-oriented AI redistributes and smooths cognitive workload across shifts and functions, fostering collaboration and improving working conditions. Robotic AI, by contrast, directly substitutes physical labor at measurable productivity multiples. While both fall under the broad label of AI in warehouse operations, the implications for workforce planning and organizational design differ markedly between planning-oriented tools and robotic systems.

4.4 Implementation Barriers and Enabling Factors

The fourth aggregate dimension addresses the organizational and technical conditions determining the successful adoption and sustainability of AI-driven decision support. Interview data revealed three primary categories of barriers and enablers: technical integration challenges, change management and organizational culture, and structural factors such as company size, governance, and cost.

4.4.1 Technical integration as a primary implementation barrier

The most frequently cited technical barrier was the difficulty of integrating AI solutions with existing Warehouse Management System (WMS) infrastructure. M. Engberg, PostNord’s previous Manager of Logistics Development, described the core problem: “We had a problem because of our WMS system. We needed to build bridges between the AI solution and the WMS,” noting that this was an ongoing issue: “every time the WMS was updated, that created problems again.” At L’Oréal, collaboration with IT was considered key for managing this complexity: “Define the processes and work hand in hand with IT” was described as the operative approach to align warehouse needs with technical implementation. The WMS was defined as the operational backbone of warehouse activity: “the WMS is the transactions that are happening inside the site,” highlighting that integration with AI tools presents a systems-related challenge with significant operational risk. The Labor Management System was also identified as a key module, with C. Percheron noting, “I am the current champion of this module.” Beyond warehouse-specific systems, Consafe described Enterprise Resource Planning (ERP) integration

as a necessary feature: “It integrates with ERP systems for articles and orders,” indicating that effective AI-supported replenishment requires connectivity across multiple system layers.

This fragility illustrates a larger challenge: AI solutions are often added to existing technology stacks rather than developed natively, resulting in dependencies that require continuous maintenance. Rather than a one-time implementation hurdle, technical integration emerges from these accounts as an ongoing operational responsibility, one that extends across WMS, ERP, and labor management systems simultaneously, and that demands sustained coordination between warehouse operations and IT functions.

4.4.2 Change management and resistance to adoption

In addition to technical barriers, organizational and behavioral resistance emerged as serious challenges to implementation. Engberg noted that current routines created friction: “It was also a challenge because people are used to working in a certain way, and then a new way comes in,” with resistance especially pronounced among experienced employees: “people with a lot of experience tend to be more rigid.” According to Unilever’s interviewee, there are clear differences between employee groups: “If you’re speaking to a warehouse team member who has decades of experience doing things a specific way, they’re more likely to feel threatened and resistant.” He emphasized that resistance to change, rather than to AI itself, is the main challenge: “The change itself would already be a significant hurdle, separate from the AI aspect entirely.” He also noted that background and digital literacy influence perceptions: “How someone perceives AI really depends on their background … The level of existing knowledge greatly influences how threatened someone feels.” In contrast, younger, digitally literate employees were described as having a different perspective: “If you’re addressing a young, digitally literate team that understands how AI can support the business, the mindset shifts completely, they see it as a tool that helps them rather than replaces them.”

In parallel, the Consafe interviewee verified this pattern, observing that “users struggle to implement new features due to time constraints and resistance to change, often believing their operations are unique.” This perception of uniqueness complicates the application of best practices and delays the adoption of applicable capabilities. At L’Oréal, company skills-development exercises were implemented to address this challenge: “We have many trainings, even mandatory

training, that employees have to follow to gain competencies in AI," showing a structured organizational effort to build capability and reduce resistance. This effort was further supported by a wider democratization effort: "There is a democratization of AI use across the organization," which extends AI literacy beyond specialist roles. Standard Operating Procedures (SOP) based chatbots were cited as a practical example: "We can load some of our SOPs or manuals into the system," thereby making operational knowledge easier to access and interactive.

This generational and experiential divide shows that change management strategies must be adapted to the workforce's backgrounds and digital competence levels. Notably, the framing from Unilever's interviewee, that resistance to change itself is the primary hurdle, independent of AI, suggests that organizations risk misdiagnosing implementation failures as technology problems when they are fundamentally human and organizational ones. The L'Oréal case illustrates one structured response to this challenge, combining mandatory training with broader AI democratization efforts to reduce resistance and build organizational readiness progressively.

4.4.3 Organizational size, governance, and competitive pressure

The pace and approach to AI adoption were reported as strongly influenced by organizational size and structural complexity. According to the Sweco consultant, "smaller or newer companies are more willing to explore it than bigger companies," while larger organizations are slowed by security and regulatory considerations: "Big companies have a lot of security issues and other things to consider." Structural fragmentation was also noted: "In a big company, you often have different cities, countries, or departments, and that creates more structure and more barriers." In contrast, new companies were described as more agile because "they build it into the company from the start." At L'Oréal, governance introduced additional complexity: "I had to defend that need with the core group," indicating that site-level AI initiatives must go through corporate governance processes before standardization. Although these challenges exist, competitive pressure was identified as a strong motivator: "If we do not do it, our competitors will," and the organization showed a clear devotion to implementation: "This year, for sure, we will implement AI." Cost was also a decisive factor in adoption decisions: "the license will increase the fee by X%," with C. Percheron adding, "we are now understanding the benefits of AI" as part of efforts to build a persuasive business case. Ongoing automation investments provided additional context: "We are investing in automation in some sites," situating AI inside broader warehouse

modernization efforts. The Sweco interviewee's observation that "I also think this is just the beginning of the AI era" highlights that current implementations, even in mature organizations, represent early stages of a longer adoption trajectory. From the solution provider perspective, data volume and maturity were identified as differentiators: "Data volume coming from more mature companies are more valuable to us as service providers, than the ones from the smaller companies," suggesting that larger organizations, while slower to adopt, ultimately provide richer data infrastructures for AI improvement.

On the other hand, Unilever's employee identified a unique barrier to AI adoption in warehouses managed by third-party logistics providers. In this setup, AI implementation requires contractual agreement: "It would need to be part of the contractual agreement, any specific software, programs, or infrastructure requirements would need to be negotiated and agreed upon." Similar to many companies, including L'Oréal, any investment would require formal financial justification: "It would need to be justified through a cost-benefit analysis."

This pattern, in which adoption depends on validated value, reflects the institutional logic of large organizations, where technology investments require measurable returns: "We know that it works and adds value." The Unilever case further illustrates that for organizations operating under outsourced logistics models, AI adoption is not only an internal decision but a multi-stakeholder negotiation, adding a layer of structural complexity absent in vertically integrated warehouse operations. Taken together, these accounts suggest that organizational size and governance structure shape not just the speed of AI adoption but the very nature of the decision-making process required to initiate it.

4.5 Emerging and Expanding AI Capabilities

The fifth aggregate dimension captures forward-looking developments and advanced AI applications identified by interviewees as either currently deployed or under active development. These findings go beyond immediate planning and replenishment to include autonomous robotic picking, AI-native warehouse management system (WMS) capabilities, and the wider expansion of AI toward orchestration and general operational management.

4.5.1 Robotic AI for autonomous order picking

The most technologically advanced application identified in the study was presented by the founder of SICS AI, whose organization has developed an AI system for robotic order picking in industrial warehouse environments. The system operates at the level of individual picking execution: "It controls robots in warehouses and industrial settings" and specifically "controls the individual pick." Its design is hardware-independent: "It is robot-agnostic, so it can control any type of robot and use any type of sensor, and one AI controls all the robots," establishing it as a platform-level solution rather than one limited to specific equipment. The system has progressed beyond proof-of-concept: "We have our first AI deployed in an industrial robot at a third-party site in Helsingborg, with amazing performance and capabilities." Performance metrics are notable: "We pick about four times faster than a human, depending on the objects," and the system's operational continuity, "We can work 24/7", removes restrictions linked to human-centered warehouse operations. The learning architecture is described as federated: "If one robot learns something on the job, the other robots also get access to that knowledge. It becomes a knowledge explosion within the system," enabling networked learning across deployment sites. The AI's learning model is characterized as emergent and experience-based: "It learns more like a child," with strong generalization capability: "Once it learns something, it can apply that in other situations." The economic implications go beyond the picking function itself: "With more efficient picking, you do not need as much storage capacity anymore. So you can save money on investment and use that money for investments in other parts of the supply chain." The founder noted that this domain is distinct from planning and forecasting AI: "I think existing AI technology can be used for that, and it is already used in some ways. But that is a different domain from where we are right now," positioning robotic AI as a separate but complementary capability. Future development is focused on orchestration rather than individual control: "Right now, we only control individual industrial robots. In the future, we will have robot orchestration functionality, but we do not have that yet."

This case illustrates a category of AI application that differs fundamentally from the decision support tools described elsewhere in this study. Rather than augmenting human planners, robotic picking AI directly substitutes physical labor at measurable performance multiples, operating continuously and learning across networked deployments. The federated

learning architecture in particular represents an emergent capability, one where competitive advantage compounds across sites over time, that has received limited attention in the existing warehouse AI literature.

4.5.2 Expanding AI capabilities within WMS

From a product development standpoint, Consafe's Product Owner described a pipeline of AI capabilities being integrated into the Astro Warehouse Management System (WMS), extending beyond replenishment to encompass workforce management, mapping, and operational analytics. Workforce prediction was identified as an emerging area: "Workforce prediction is another AI area," "enhanced by external sales forecasts from partners" to facilitate anticipatory staffing rather than reactive scheduling. AI-driven warehouse mapping was described as a near-term development: "We're developing AI for automated mapping, expected in 1-2 years," said Hylén, which would automate the complex, multi-constraint slotting process that currently requires manual configuration. The existing system already incorporates article association logic for placement optimization: "Current AI includes article associations for optimal placement," leveraging co-purchase relationships to minimize picker travel distance. Optimization of empty forklift movement was also noted: "We optimize empty forklift moves," addressing inefficiencies in heavy-storage warehouse environments. The broader system architecture supports hybrid operational environments: "It supports both manual and automated warehouses," and integrates with automation at multiple levels of control, ranging from full white box control, where the WMS directs all equipment actions, to black box integration, where automation operates independently and the WMS coordinates only pallet flows. The machine learning layer forms the base of these adaptive features: "Most features in Astro use machine learning," establishing machine learning as the technical core rather than a supplementary component.

These accounts suggest that AI capabilities within WMS platforms are expanding along two parallel trajectories: deeper integration into existing operational functions such as slotting and forklift routing, and extension into new domains such as workforce prediction and automated mapping. The positioning of machine learning as the technical core, rather than a supplementary add-on, signals a broader architectural shift in how WMS platforms are being designed, with adaptive intelligence embedded at the foundation rather than layered on top.

4.5.3 Expanding AI ecosystem strategy

L'Oréal's interviewee described the organization's transition from isolated AI tools to a diversified and institutionalized AI ecosystem. The current environment includes multiple generative AI platforms: "We call it L'OréalGPT, but we also have Gemini and Claude," showing a careful strategy of diversification rather than dependence on a single vendor. The organization's leadership in AI adoption within the supply chain is demonstrated by its participation in an advanced technology partnership with other companies who "offer the possibility to some companies to become a beta tester. And on the AI effort, we pushed to be included in this group," with Manhattan identified as the primary technology provider. In addition to that, existing forecasting infrastructure acts as a foundation for AI integration: "We have some software that we are using to estimate forecast based on data," and the global replenishment process follows a structured bottom-up approach: "there is a global process, and it is bottom-up," where demand planning at the country level consolidates upward to inform factory production planning: "The factories receive the information through the process and then transforms it into a production plan." Distribution center operations are also aligned with this global flow: "The IDC follows the production of the factories," with quarantine and release management conducted outside the factory to maintain production continuity: "We have optimized the process so that quarantine does not happen in the factory." AI's role within this architecture is expected to expand significantly: "This year, for sure, we will implement AI," supported by a mandatory training program and a multi-model AI environment. The L'Oréal case further demonstrates how AI-supported replenishment enhances supply chain resilience: "For supplier delays, the system simulates the impact on inventory and suggests mitigation actions, like increasing orders from alternative suppliers or reallocating stock between warehouses," moving beyond passive reporting to active scenario modeling and mitigation recommendations.

The operational value of simulation capabilities is especially clear when compared to organizations that lack them. For Unilever UAE, the management of port congestion, border delays, and rising costs across Arabian markets is exclusively done through manual intervention and experience, without system support for scenario modeling or mitigation: "We did experience some disruptions with outbound shipments across the Arabian markets, port congestion, border delays, and rising costs […] despite all of that, we delivered well and actually achieved good growth in Q1."

Although the outcome was positive, the absence of predictive simulation tools meant resilience relied entirely on individual experience and judgment.

Together, these two cases illustrate the gap that emerging AI capabilities in disruption management are meant to fill. The contrast between L'Oréal's active scenario modeling and Unilever UAE's experience-driven manual resilience is particularly enlightening, as it shows that positive operational outcomes are achievable without AI, but that the cognitive load and organizational risk of sustaining them are substantially higher. This gap between AI-enabled and non-AI-enabled resilience represents one of the more concrete and empirically grounded contributions of this study.

5 Discussion

This chapter discusses the findings in relation to the literature review and the research questions. While the findings show that AI-driven decision support is increasingly present in warehouse operations, they also indicate that its value is not mainly created through full automation. Instead, AI appears to support warehouse decision-making by improving visibility, generating recommendations, identifying exceptions, and enabling more adaptive planning. The discussion is structured around five central interpretations: AI as augmentation rather than automation, AI-supported adaptive planning for warehousing, operational value creation, socio-technical conditions for effective use, and the emerging movement from decision support toward automation and orchestration.

5.1 AI as Augmentation Rather Than Automation

Previous research on AI in organizational decision-making emphasizes that AI should be viewed not only as a substitute for human decision-makers but also as a tool that complements human judgment. Berretta et al. (2023) describe this relationship as human-AI symbiosis, in which AI provides analytical capacity and humans offer contextual understanding and judgment. Similarly, Raisch and Krakowski (2021) discuss the automation–augmentation paradox, suggesting that while AI may automate certain tasks, it also increases the importance of human oversight, interpretation, and exception handling. Trust remains central in these arrangements, since users must judge when to rely on or question automated recommendations (Lee & See, 2004).

The findings of this study reinforce the perspective that AI-driven decision support is primarily regarded as a means of augmenting, rather than replacing, human decision-making. Interview respondents consistently emphasized that AI supports daily work, generates recommendations, and helps planners focus their attention, while final decision authority remains with humans. This perspective was clear even in organizations that had not yet implemented AI in warehouse planning, where respondents still described AI as a potential tool for visibility and decision support rather than as a replacement for experienced planners. These observations emphasize the continued importance of human expertise across varying levels of AI maturity. This aligns closely with existing literature on human-AI decision-making, which suggests that AI can enhance organizational decision processes, while human judgment remains essential for contextual

interpretation, evaluation, and responsibility (Jarrahi, 2018). It also exemplifies the automation–augmentation paradox described by Raisch and Krakowski (2021), in which AI may automate certain tasks while simultaneously increasing the importance of human oversight and interpretation.

Evidence from the interviews suggests that the logic of augmentation is especially evident in warehouse replenishment. AI systems can generate replenishment suggestions, update forecasts, or flag operational exceptions; however, planners continue to review, adjust, and validate these outputs. This suggests that AI's primary contribution is not the elimination of human involvement, but rather the transformation of human work. Planners shift from performing repetitive calculations and routine order processing to focusing on exception management, validation, and cross-functional coordination. Thus, AI-supported warehouse planning emerges as a socio-technical process in which technology and human expertise are interdependent.

Therefore, this outcome contributes to the literature on AI in supply chain management by illustrating how augmentation shows in practical warehouse settings. While existing research often highlights AI's analytical potential, the observed evidence suggests that organizations do not automatically convert these capabilities into autonomous decision-making. Instead, the application of AI is constrained by factors such as trust, accountability, operational complexity, and the necessity for human contextual knowledge. This is particularly relevant in replenishment decisions, where trade-offs among service levels, inventory costs, supplier constraints, and disruptions are not always fully addressed by automated recommendations.

The study's data further underscore the importance of appropriate system reliance. Respondents described cases where users either placed excessive trust in the system or did not trust it sufficiently. This observation supports Lee and See's (2004) argument that trust in automation must be carefully regulated. Insufficient trust may prevent organizations from realizing the benefits of AI recommendations, while excessive trust can result in uncritical acceptance of outputs. In the context of warehouse replenishment, appropriate reliance requires both technical accuracy and user understanding. Explainability is therefore crucial, as planners are more likely to trust and utilize AI recommendations when they comprehend the reasoning behind suggested actions.

In summary, the study indicates that AI-driven decision support in warehouses is most effectively conceptualized as a form of human-AI coordination. The value of AI is found in enhancing the quality and visibility of decision inputs, while human planners retain responsibility for interpreting recommendations, managing exceptions, and making final decisions. These findings address the second research question by demonstrating that effective AI utilization depends not only on technical capability but also on users' understanding, trust, and interaction with AI outputs.

5.2 AI-Supported Adaptive Planning for Warehousing

The literature on AI-enabled supply chain planning differentiates between predictive analytics, prescriptive analytics, and real-time adaptive decision support. Predictive analytics facilitates forecasting by identifying patterns in historical and operational data, while prescriptive analytics converts these predictions into recommended actions (Lepenioti et al., 2020). In supply chain contexts, these capabilities are critical because planning decisions must address demand volatility, supplier uncertainty, and evolving operational constraints (Toorajipour et al., 2021). Traditional planning approaches are often insufficient in such environments, as they depend on stable assumptions, fixed rules, or human judgment under uncertainty (Perera et al., 2019).

The findings strengthen existing literature by demonstrating that AI-supported replenishment transforms warehouse planning from static, routine decision-making to more adaptive and exception-based approaches. Traditional methods, including rule-based systems and statistical forecasting, are constrained when supply chain conditions are volatile or when assumptions regarding demand and lead times shift. The literature identifies predictive, prescriptive, and real-time adaptive mechanisms as essential AI capabilities for supporting supply chain planning under uncertainty. The present findings validate this perspective by illustrating how AI is employed to generate replenishment proposals, forecast demand, update safety stock, identify exceptions, and facilitate warehouse slotting and routing.

A significant discovery is that lead time uncertainty has emerged as a critical application area for AI. This is significant because most of the discussion around AI in replenishment emphasizes demand forecasting, whereas the interviews indicate that lead time validation and transport uncertainty are equally important. In practice, replenishment decisions depend not only on anticipated demand but also on the timing of stock availability. Extended transport routes, supplier

delays, and discrepancies between planned and actual lead times introduce uncertainty that impacts stock availability and service levels. AI can enhance replenishment by comparing planned lead times with operational outcomes and identifying deviations before they result in shortages or excess inventory.

Additionally, respondents reported that AI-supported replenishment is not a constant practice. Solution providers described various replenishment logics, such as order-driven and low-watermark. This diversity suggests that AI-supported replenishment must be tailored to specific operational contexts. For instance, order-driven replenishment is appropriate when demand is already confirmed, while predictive replenishment is advantageous when demand patterns can be forecasted and labor resources can be planned in advance. These observations reinforce that AI tools are not universally applicable; their effectiveness depends on alignment with warehouse processes, demand characteristics, SKU complexity, and system architecture. This diversity in replenishment logics therefore implies that AI implementation strategies cannot be standardized across organizations; rather, solution design must begin with a clear understanding of the dominant replenishment logic in use, as this determines which AI capabilities are relevant, how recommendations should be generated, and at what point human validation is most critical.

Another significant contribution concerns exception management. AI systems not only generate plans but also assist in identifying potential plan failures. Interviewees described exception flags related to demand spikes, missing picking locations, late trucks, supplier delays, and operational readiness. These findings indicate that AI can enhance resilience by making deviations visible at an earlier stage. Rather than replacing planners, AI directs attention to situations where human intervention is most critical. This finding connects predictive and real-time AI mechanisms to practical warehouse decision-making by demonstrating their role in supporting proactive intervention.

Overall, AI-supported replenishment represents a shift from periodic, manual planning to more continuous, adaptive, and exception-based decision-making. This addresses the first research question by illustrating how AI enhances operational performance and resilience through improved forecasting, lead time visibility, safety stock adjustment, and exception detection. Nevertheless,

the realization of these benefits depends on effective integration with warehouse processes and the capacity of users to interpret and act upon AI-generated outputs.

5.3 Operational Value: Performance, Resilience, and Sustainability

The existing literature highlights that replenishment and inventory decisions significantly affect service levels, inventory costs, lead times, working capital, and overall supply chain responsiveness (Chopra & Meindl, 2022; Christopher, 2016). From the perspective of resilience, the capacity to respond effectively to volatility and disruptions is considered a core supply chain capability (Yu et al., 2019). Artificial intelligence (AI) in supply chain management is frequently examined as a means to enhance decision quality and responsiveness by processing complex datasets and supporting more informed planning decisions (Toorajipour et al., 2021).

The study validates this perspective by demonstrating that AI-driven decision support generates operational value primarily through performance enhancements. Respondents reported improvements in service levels, reductions in urgent shortages and excess inventory, optimized order quantities, increased picking efficiency, and more effective workload distribution. These outcomes are consistent with existing supply chain literature, which proposes that enhanced planning and decision support can elevate service levels, decrease inventory-related costs, and improve responsiveness. Specifically, in warehouse replenishment, AI contributes to performance by enabling more accurate forecasts, generating better order proposals, and facilitating timely identification of operational issues.

The most immediate performance impacts target inventory and service levels. AI-supported replenishment enables organizations to align inventory more precisely with demand, especially in volatile scenarios such as promotional periods, seasonal fluctuations, or environments with a high number of stock-keeping units (SKUs). By producing SKU-level forecasts and replenishment proposals, AI reduces the manual planning burden and allows planners to oversee a larger volume of items. This indicates that AI enhances planning scalability, enabling organizations to manage increased complexity without a corresponding rise in planning staff. However, the interview outcomes reveal that workload is not only reduced but rather redistributed. Planners devote less time to manual calculations and more to reviewing exceptions, validating AI-generated recommendations, and coordinating with other organizational functions.

The findings further indicate that AI contributes to supply chain resilience, primarily through indirect mechanisms. This is mainly achieved through enhanced visibility, accelerated detection of disruptions, and improved simulation of supplier delays or demand fluctuations support resilience. AI tools facilitate earlier risk identification and can recommend mitigation strategies such as alternative sourcing or stock reallocation. These results reinforce the perspective that AI can strengthen supply chain responsiveness under conditions of uncertainty. Nonetheless, the findings also demonstrate that extreme disruptions continue to necessitate human intervention and expertise. Consequently, AI should be regarded not as a substitute for resilience capabilities, but as a tool that augments the informational foundation for resilient decision-making.

Sustainability benefits appear to be present, but mainly as indirect outcomes of improved planning. Respondents associated AI-supported replenishment with reductions in waste, particularly for perishable or expiry-sensitive products. Improved forecasting and lead time accuracy help minimize overstocking and prevent products from becoming unsellable. Additionally, transport consolidation and a decrease in urgent replenishment orders contribute to lower costs and reduced emissions. AI applications in robotics and automation may further enhance resource efficiency by increasing picking productivity and minimizing unnecessary storage requirements. However, these sustainability outcomes should be interpreted with caution. The insights indicate that sustainability is not consistently a primary objective of AI implementation; rather, it emerges as a secondary benefit when AI enhances inventory accuracy, resource utilization, and process efficiency.

On the other hand, the findings also nuance the sustainability discussion by showing that AI's environmental contribution should not be treated as automatically positive. While several respondents linked AI-supported planning to reduced waste, fewer urgent shipments, and improved resource efficiency, one interviewee interestingly warned against selectively emphasizing efficiency gains while overlooking the energy and computational costs of AI-enabled systems. This suggests that sustainability benefits from AI-supported warehouse planning are conditional rather than natural. AI may support sustainability when it reduces expired inventory, unnecessary transport, or inefficient resource use, but these benefits need to be weighed against the broader environmental footprint of AI implementation itself.

That being said, this has an important implication for the first research question as the findings suggest that the most immediate effects of AI-driven decision support are visible in operational performance, such as service levels, workload distribution, inventory control, and lead time predictability. Resilience benefits are also evident, but mainly through improved disruption visibility, supplier-delay simulation, and faster response to exceptions. Sustainability benefits appear more indirectly through waste reduction, transport consolidation, and more efficient resource use; while accounting for the environmental footprint of AI implementation previously mentioned.

Therefore, the effect of AI is layered; it first improves decision inputs and operational coordination, and these improvements may then translate into resilience and sustainability outcomes. The operational value of AI should therefore not be understood only through efficiency metrics, but also through how work is redistributed between planners, warehouse employees, and automated systems.

5.4 Preconditions: Data, Integration, Trust, and Organizational Readiness

Existing research on AI readiness demonstrates that successful AI implementation relies not only on technological capabilities but also on organizational factors such as data availability, workforce skills, governance structures, strategic alignment, and integration with current processes (Jöhnk et al., 2021). Similarly, literature on smart warehousing sheds light on the necessity of addressing human, technical, and organizational preparedness in addition to technical functionality (Mahroof, 2019). This socio-technical perspective is especially relevant to warehouse replenishment, where AI-generated recommendations must be integrated with WMS/ERP systems, trusted by end users, and incorporated into routine planning activities.

The analysis provides strong support for that socio-technical perspective. A key contribution of this study is the identification that AI effectiveness is dependent on a comprehensive configuration of data, systems, users, and organizational routines, rather than the AI tool alone. Respondents consistently highlighted that data quality, WMS/ERP integration, user trust, training, organizational readiness, and cost justification are critical factors of effective AI utilization. These results directly address the second research question regarding the organizational and technical

factors influencing effective use. The evidence indicates that AI implementation should not be viewed solely as a technological enhancement; rather, it necessitates alignment among systems, personnel, processes, and governance structures.

Data quality was identified as a critical technical precondition. Respondents repeatedly noted that inadequate master data results in unreliable AI outputs, reinforcing the concept that AI systems magnify the quality of their inputs rather than compensate for deficient data foundations. In the context of warehouse replenishment, where decisions rely on SKU data, stock levels, supplier lead times, order histories, promotional information, and warehouse constraints, unreliable data can compromise both output accuracy and user trust. Consequently, organizations aiming to implement AI-supported replenishment should prioritize investments in master data governance and data integration, while ensuring fair working conditions for data gatherers and labelers.

Additionally, the results further indicate that data quality constitutes not only a continuous post-implementation challenge but also a prerequisite for AI adoption. Organizations that have not yet implemented AI may still encounter fragmented or manually interpreted data, which limits their capacity to transition toward AI-supported decision-making.

System integration emerged as a significant barrier. Interviewees reported challenges in connecting AI tools to existing WMS environments and issues arising from system updates. This is particularly important because warehouse decision support relies on operational systems that manage transactions, picking, replenishment assignments, and inventory movements. Without integration, AI-generated recommendations may remain isolated from execution. The findings suggest that the value of AI is determined not only by algorithmic sophistication but also by the extent to which AI tools are embedded within the operational technology stack.

Trust and explainability were also identified as critical factors. As previously mentioned in section 5.1, interviews revealed that users may either over-rely on or distrust AI, both of which diminish effective utilization. Explainability emerged as a key area for improvement, as users require an understanding of the logic behind AI-generated recommendations. In warehouse replenishment, explainability is especially important because planners must justify decisions that impact stock availability, costs, and service levels.

Organizational readiness and change management were also found to influence AI adoption. The findings indicate that larger, more mature organizations often possess richer data resources but may experience slower progress due to governance requirements, security concerns, operational complexity, and resistance to change. In contrast, smaller or newer firms may demonstrate greater flexibility and willingness to experiment yet may lack the scale or data maturity necessary to fully leverage AI. This dynamic creates a tension between organizational agility and data richness. Moreover, training, democratization of AI knowledge, and development of robust business cases were identified as essential enablers. In the absence of user capability and organizational justification, AI initiatives are likely to remain at the pilot stage rather than becoming integrated into routine decision-making.

In summary, effective AI-supported replenishment is contingent upon both technical and organizational conditions. Data quality, system integration, trust, explainability, training, governance, and cost justification are fundamental determinants of AI value generation, rather than secondary implementation details. These findings suggest that the success of AI in warehouse environments is shaped not solely by technological factors, but by the organization's capacity to establish conditions that enable users to rely on, interpret, and act upon AI-generated recommendations. Therefore, in relation to the second research question, the outcomes indicate that the effective use of AI-driven decision support is shaped by a combination of technical foundations, such as data quality and system integration, and organizational foundations, such as trust, training, governance, and change readiness.

5.5 From Decision Support Toward Automation and Orchestration

The following themes go beyond the primary focus on AI-supported warehouse replenishment and are therefore presented as emerging findings rather than as direct responses to the research questions. These themes are included because multiple respondents indicated that AI decision support is likely to become increasingly integrated with warehouse execution, robotics, and broader automation.

The smart warehousing literature indicates that warehouse digitalization increasingly integrates information systems, automation technologies, and human labor within unified operational environments (Mahroof, 2019). At the same time, the automation–augmentation paradox

demonstrates that artificial intelligence (AI) can both support human decision-making and automate specific operational tasks, resulting in distinct implications for roles, required skills, and organizational structures (Raisch & Krakowski, 2021). Within supply chain management, AI is not confined to planning support; it also facilitates broader operational transformation (Toorajipour et al., 2021).

The data acquired align with this broader trend. The final discussion point addresses the expanding scope of AI within warehouse operations. Although the primary focus of this thesis is AI-driven decision support, the results demonstrate that AI capabilities are advancing toward physical automation and orchestration. Notable examples include robotic order picking, AI-enabled warehouse mapping, workforce prediction, route optimization, order batching, and prospective robot orchestration. These developments suggest that AI in warehouses is no longer restricted to planning recommendations; it is increasingly involved in operational execution and automated control. While these emerging capabilities extend beyond the thesis's specific focus on replenishment, they are significant as they illustrate the growing connection between warehouse decision support, execution, automation, and orchestration.

This progression introduces a critical distinction between decision-support AI and automation-oriented AI. Decision-support AI assists planners by enhancing forecasts, generating recommendations, and increasing exception visibility. In contrast, automation-oriented AI can directly control robotic systems leading to non-human assisted decisions, optimize movement, or replace manual picking tasks. The findings indicate that while these two approaches are interconnected, they are not identical.

For instance, robotic picking enhances operational transparency and produces data that can inform forecasting, yet it remains distinct from replenishment planning AI. This distinction is significant because the organizational implications differ; decision-support AI primarily alters planning roles, whereas automation-oriented AI may directly influence warehouse labor and job design.

The shift toward orchestration further implies that future warehouse AI will likely achieve greater integration across planning and execution functions. If AI systems are capable of coordinating multiple robots, optimizing workforce allocation, adjusting replenishment flows, and integrating with warehouse management systems (WMS) and enterprise resource planning (ERP) systems,

the distinction between planning support and operational automation may become less defined. This observation aligns with the broader literature on AI and supply chain digitalization, which emphasizes that value is generated not only by individual tools but also through interconnected systems.

Simultaneously, this expansion introduces ethical and managerial considerations. Although automation can enhance speed, productivity, and resource efficiency, it may also displace manual labor or alter skill requirements. The findings therefore indicate that AI implementation should be assessed not only by operational performance but also by its effects on the workforce, transparency, and responsible use. This consideration is particularly important, as the thesis demonstrates that AI represents both a technical capability and a socio-technical transformation of warehouse work.

In summary, the findings demonstrate that AI in warehouses is progressing along two interconnected trajectories; the augmentation of planning decisions and the automation of operational execution. While the present value of AI-supported replenishment is primarily found in decision support and exception management, emerging capabilities suggest that future warehouse AI will increasingly facilitate integrated orchestration of personnel, systems, inventory, and machinery. While this development extends beyond the primary focus on replenishment decision support, it is relevant to the second research question as it highlights the organizational and technical conditions required for effective AI integration in warehouse operations.

Although automation, robotics, and AI ecosystems are pertinent to the future development of AI in warehouse environments, the main contribution of this thesis is the examination of AI-driven decision support for replenishment. This focus includes its recognized impact on operational outcomes as well as the organizational and technical factors that influence effective implementation.

5.6 Summary of Discussion

The discussion demonstrates that AI-driven decision support influences warehouse replenishment primarily by enhancing decision inputs, visibility, and responsiveness, rather than replacing human decision-makers. AI assists planners by generating replenishment proposals, validating

assumptions, identifying exceptions, and enabling more adaptive planning. Its operational value is most evident in improved service levels, inventory control, workload redistribution, productivity, and lead time management. Resilience and sustainability benefits are also observed, although these are more indirect and emerge through improved disruption response, waste reduction, transport consolidation, and more efficient resource utilization. However, sustainability benefits should be understood as conditional rather than automatic, as the efficiency gains of AI-supported planning must be weighed against the energy and computational costs associated with AI implementation itself.

The discussion further indicates that the effective use of AI depends on several organizational and technical preconditions. Factors such as data quality, integration with Warehouse Management Systems (WMS) and Enterprise Resource Planning (ERP), explainability, user trust, training, change management, governance, and cost justification all influence whether AI can transition from technical capability to practical value. Consequently, AI-supported warehouse replenishment should be conceptualized as a socio-technical process involving the interaction of technology, human judgment, and organizational context.

Collectively, the study advances the understanding of AI-driven decision support by demonstrating that its value in warehouse replenishment does not primarily stem from full automation. Instead, value arises through augmentation, exception-based planning, enhanced visibility, and improved coordination between human planners and AI-enabled systems. Thus, the discussion links the empirical findings to existing literature by establishing that AI-supported warehouse replenishment should be regarded as a socio-technical decision-support process rather than a purely technological or fully automated solution.

6 Conclusion

This thesis examines the application of AI-driven decision support in warehouse operations, focusing specifically on replenishment-related decision-making. The research seeks to determine how AI-supported tools affect operational performance, resilience, and sustainability, and to identify the organizational and technical factors that influence their effective implementation. Drawing on qualitative interviews with practitioners and solution providers, as well as insights from conferences and site visits, the findings indicate that AI in warehouse replenishment functions primarily as a decision-support capability rather than as a substitute for human planners. Overall, the study demonstrates that AI creates value by enhancing decision inputs, increasing operational visibility, generating replenishment proposals, identifying exceptions, and facilitating more adaptive planning. However, the realization of this value is highly contingent upon data quality, integration with existing systems, user trust, organizational readiness, and the capacity of planners to interpret and implement AI-generated recommendations.

6.1 Answer to the Research Questions

6.1.1 Answer to RQ1: Influence on Operational Performance, Resilience, and Sustainability

The first research question asked:

How does the implementation of AI-driven decision support for warehouse replenishment influence operational performance, resilience, and sustainability?

The results show that AI-driven decision support most directly impacts operational performance. AI-supported tools boost service levels, reduce excess inventory, optimize order quantities, improve lead time predictability, and shift workload from routine manual planning to exception management. In warehouse environments marked by high SKU complexity, demand volatility and frequent operational constraints, AI enables planners to process larger data volumes and concentrate on deviations needing human intervention.

AI contributes to organizational resilience, though this effect is primarily indirect. The findings demonstrate that AI improves disruption visibility by detecting exceptions, simulating supplier-

delay impacts, identifying shipment risks, and enabling faster responses to demand fluctuations. Instead of eliminating uncertainty, AI allows organizations to identify and address uncertainty at an earlier stage. Consequently, AI strengthens resilience mainly by enhancing the informational foundation for decision-making and supporting proactive intervention.

The impact of AI on sustainability is also predominantly indirect. Interviewees associated AI-supported replenishment with reduced waste, lower expiry risk, transport consolidation, fewer urgent replenishment orders and more productive resource usage. However, the study shows that sustainability benefits are not guaranteed. AI systems may incur considerable computational and energy costs, so their environmental contribution depends on whether process efficiency gains exceed the additional resource demands of implementation. Thus, AI can only support sustainability when implemented in ways that clearly reduce waste, unnecessary transport and inefficient resource use.

6.1.2 Answer to RQ2: Organizational and Technical Factors Shaping Effective Use

The second research question asked:

What organizational and technical factors shape its effective use?

The results show that the effective use of artificial intelligence is determined by a set of technical and organizational factors. From a technical perspective, master data quality, system integration, real-time data availability, and system reliability are critical. In warehouse environments, this typically requires the combination of AI-generated recommendations with existing operational systems, such as Warehouse Management Systems (WMS) or Enterprise Resource Planning (ERP) platforms. AI-generated recommendations provide value only when they are based on accurate, integrated data and can be seamlessly connected with the systems responsible for warehouse planning and execution.

From an organizational standpoint, effective implementation depends on user trust, explainability, training, change management, governance, and cost justification. Planners must be able to determine when AI recommendations should be accepted, modified or challenged. The study shows that both overdependence and skepticism can reduce the value of AI. Therefore, organizations should foster appropriate reliance by providing training, ensuring explainable

outputs and formulating clear boundaries of responsibility between human planners and AI systems.

Taken together, the answer to both research questions is that AI-driven decision support enhances warehouse replenishment not through complete automation, but through augmentation. The value of AI arises when it increases the visibility, speed, and quality of planning decisions, while human planners retain responsibility for contextual judgment, exception management, and final decision-making.

6.2 Theoretical Contribution

This thesis contributes to the literature on artificial intelligence (AI) in supply chain management by redirecting focus from algorithmic potential to practical application in warehouse decision-making. While existing research frequently highlights AI's capacity to forecast, optimize, or automate supply chain decisions, this study provides empirical insight into how practitioners interpret and apply these capabilities within actual warehouse and replenishment settings.

The primary theoretical contribution is the conceptualization of AI-supported warehouse replenishment as a socio-technical decision support process. The results show that the value of AI emerges from the interaction between analytical systems, human judgment, operational routines and organizational conditions. Rather than simply replacing existing planning processes, AI transforms the nature of planning work by shifting planners' roles toward exception management, validation, and cross-functional coordination.

The study further shows that AI-supported replenishment covers a range of distinct practices. These include lead time validation, replenishment proposal generation, predictive safety stock adjustment, exception flagging, slotting optimization, warehouse management system (WMS)-based replenishment logic, and emerging automation and orchestration capabilities. This broader perspective improves understanding of AI in warehouse operations by showing its support for both planning and execution related processes.

6.3 Managerial Implications

The findings present several consequences for managers who seek to implement AI-driven decision support systems in warehouse replenishment. Instead of reiterating recognized best practices, these implications are based on patterns that are less frequently documented in existing literature, including unexpected results regarding use case prioritization, organizational readiness, and indirect effects of certain AI-driven benefits.

First, lead time uncertainty, rather than demand forecasting, should be prioritized as the primary use case for AI adoption.

Although existing literature primarily emphasizes AI's value in demand sensing for replenishment, the results show that the differences between planned and actual lead times represent the most urgent and major planning failure. Organizations should initiate AI adoption by addressing lead time validation and transport uncertainty, since these factors directly compromise the reliability of replenishment decisions, regardless of forecast accuracy.

Second, replenishment logic must be diagnosed before selecting AI capabilities.

The findings show that organizations employ fundamentally different replenishment logics, including order-driven, low-watermark, and predictive approaches. AI tools deliver value only when aligned with the dominant logic. Therefore, managers should conduct a replenishment logic assessment as the initial step in AI scoping, rather than approaching AI implementation as a generic technology deployment.

Third, workload redistribution, not workload reduction, should be expected organizational outcome.

A common managerial assumption is that AI reduces the planning headcount. However, evidence consistently demonstrates the opposite: AI reallocates planners from routine calculations to exception management, supplier coordination and validation activities. Organizations that position AI primarily as a headcount reduction tool run the risk of creating misaligned expectations, resistance and governance challenges. Success metrics should instead focus on decision quality, exception response time and planner capacity utilization.

Fourth, explainability is a functional prerequisite, not a desirable feature.

The findings reveal that both overreliance and underutilization of AI can occur simultaneously within the same organizations, with poor explainability identified as the root cause in both cases. Managers should require AI vendors to demonstrate explainability at the recommendation level, not only at the model level, prior to deployment.

Fifth, sustainability and resilience benefits should be treated as second-order results dependent on planning accuracy.

The results show that waste reduction, transport optimization and improved disruption response do not result from AI systems designed with sustainability or resilience objectives. Instead, these outcomes arise as downstream consequences of more accurate replenishment decisions. Reduced overstock generates less waste, more predictable ordering enables transport consolidation, and faster exception detection improves disruption response. Managers should therefore avoid framing AI adoption as a sustainability initiative. Sustainability and resilience gains are best understood as lagging indicators of planning performance. If replenishment accuracy improves, these outcomes tend to follow. Setting sustainability KPIs as primary success criteria for AI implementation risks misattributing outcomes and confusing the actual mechanism through which AI creates value in warehouse operations.

Collectively, these findings indicate that successful AI adoption in warehouse replenishment depends more on organizational sequencing than on technological capability. The evidence demonstrates a consistent pattern: organizations that align AI deployment with existing replenishment processes, prioritize explainability, and redefine expected outcomes are more likely to achieve sustained operational value. Managers who approach AI implementation as a simple technological addition, rather than as a socio-technical transition, are likely to face avoidable challenges that remain underexplored in most current literature.

6.4 Limitations

This study presents several limitations. The research draws upon a limited number of interviews and does not seek statistical generalization. Instead, the findings offer analytical insights into the perception and use of AI-driven decision support within selected warehouse and supply chain contexts.

The empirical material also includes both users of AI-supported tools and solution providers. This approach offers a broader perspective, but it also means that some findings reflect provider-side expectations and intended system capabilities rather than solely user-side experiences. Future research could examine one or several user organizations in greater depth over an extended period. Another limitation is the reliance primarily on interview-based perceptions rather than direct measurement of operational key performance indicators (KPIs). Although respondents reported improvements in service levels, workload, lead-time predictability, waste reduction, and efficiency, these effects were not independently verified through quantitative performance data.
Finally, the participating organizations vary in industry, size, AI maturity, and warehouse configuration. This diversity enriches the analysis but complicates direct comparisons between cases. Consequently, the findings identify patterns across contexts rather than evaluating a single standardized AI implementation.

6.5 Future Research

Future research may extend this study in several directions. Quantitative studies could evaluate the impact of AI-supported replenishment on key performance indicators such as service levels, inventory turnover, stockouts, waste, lead times, and transport efficiency. Such analyses would complement the qualitative insights presented in this thesis.

AI implementation could be studied over time, from pilot stages to full operational deployment, could provide insights into the development of trust, data quality, user acceptance, and organizational routines throughout the implementation process.

Comparative studies across industries could investigate whether the effectiveness of AI-supported replenishment varies according to demand patterns, product characteristics, and warehouse structures. For instance, fast-moving consumer goods, pharmaceuticals, retail, construction, and e-commerce sectors may encounter distinct benefits and barriers.

Further research is required to examine the sustainability implications of AI in warehouse operations. Although AI has the potential to reduce waste and enhance resource efficiency, its computational and energy demands must also be considered. Future studies should assess whether AI-supported warehouse systems produce net environmental benefits.

Finally, future research could investigate the boundary between decision support and automation. As AI capabilities progress from planning recommendations to robotic picking, workforce allocation, and warehouse orchestration, it is important to investigate how these advancements influence planners, warehouse workers, managerial control, and the organization of warehouse work.

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8 Appendix

8.1 Appendix: Interview Guide

8.1.1 Interview Questions (Semi-Structured): Users

1. How would you describe your role in the warehouse, and what key decisions are you responsible for?

2. How do you use AI-driven or advanced decision support tools when making replenishment or inventory planning decisions

3. In what ways do these tools support replenishment?

4. How has the use of AI decision support influenced inventory performance, such as service levels, inventory, lead times, or workload?

5. How has AI decision support affected your ability to respond to disruptions, such as demand fluctuations or supplier delays?

6. How, if at all, has AI-supported replenishment or inventory planning influenced resource efficiency or sustainability in the warehouse?

7. How has the introduction of AI tools changed your role, responsibilities, or ways of working?

8. What affects how useful AI is for replenishment decisions, and what challenges or improvements have you seen?

8.1.2 Interview Questions (Semi-Structured): Solution Providers

1. How would you describe your role in designing or advising on warehouse and replenishment solutions, and what key decisions are you responsible for?
2. How do you design or implement AI-driven decision support tools for replenishment or inventory planning?
3. How are the AI tools you design intended to support replenishment decisions in practice?
4. Based on your projects, how do AI-supported solutions impact inventory performance (e.g., service levels, inventory levels, lead times, workload)?
5. How do the solutions you design help organizations respond to disruptions like demand fluctuations or supplier delays?
6. How do AI-driven replenishment solutions contribute to resource efficiency or sustainability in warehouse operations?
7. How do these AI tools typically change the roles, responsibilities, or ways of working for warehouse staff and planners?
8. What factors influence how effective AI is in replenishment decisions, and what common challenges or areas for improvement have you observed across implementations?
9. To what extent do the solutions you design get used as intended, and where do you typically see gaps between design and actual usage?

8.2 Coding Table

Interviewee	Company	Position	Concept Number	First-Order Concepts	Evidence	Interpretation
Anders Sjögren	Astrazeneca	Associate Director Global Supply Chain Process	1	AI is used indirectly, not as a core decision-maker	"We do use AI indirectly, tools like Copilot or our internal ChatGPT, but they mainly support daily tasks rather than core supply chain decisions"	AI currently supports general productivity rather than driving replenishment decisions directly
			2	Replenishment lead time is the key use case	"We also have another use case with Era that is further along This relates to replenishment lead time"	The interviewee identifies replenishment lead time as their most advanced AI application
			3	Lead time is a central planning variable	"replenishment lead time, the time from when we start planning a product to when it is available in the market"	Lead time spans entire process from planning to stock availability, core to replenishment
			4	Transportation lead time matters	"Part of that lead time is transportation For example, if we produce in Sweden and ship to Taiwan by boat, it might take around six weeks"	Transportation forms significant portion of total replenishment lead time
			5	Long transport routes create uncertainty	"if we produce in Sweden and ship to Taiwan by boat, it might take around six weeks"	International routes introduce significant variability in planning
			6	AI could help verify lead time assumptions	"We've introduced an AI tool to validate whether our planned lead times match reality"	AI specifically targets gap between designed/planned vs actual lead times
			7	Master data is crucial	"a key factor is having high-quality master data If your data isn't reliable, applying AI won't help; it will just produce unreliable outputs"	Data quality determines AI effectiveness—bad data produces unreliable outputs
			8	AI may help reduce waste	"we can sometimes redirect products to markets with lower requirements However, if a product can't be sold, it must be destroyed"	Better lead time accuracy could reduce unsellable inventory nearing expiry
			9	AI is not replacing human decision-making	"they mainly support daily tasks rather than core supply chain decisions So far, there hasn't been a direct impact"	Human oversight remains essential for supply chain decision authority
			10	AI is a supporting tool, not a full solution	"AI is not a magic solution, it won't fix everything"	AI augments rather than automates complex supply chain processes
			11	Planning is complex and parameter-heavy	"There are many parameters involved… balance the risk of unsold products with production efficiency"	Replenishment involves multiple interdependent variables requiring human judgment
Anonymous Interviewee 1	Anonymous Company 1	Warehouse Manager	12	Balances availability with cost	"A big part of the job is balancing availability with cost. We can't afford stockouts because that delays projects, but overstocking ties up capital and space."	Navigates trade-offs between service levels and financial efficiency
			13	Integrates diverse data	"We use an AI-based planning system that integrates historical consumption data, project timelines, and supplier lead times."	Pulls real-time and projected data for holistic planning
			14	Generates order suggestions	"It generates replenishment suggestions, basically recommended order quantities and timing for each SKU."	Provides actionable, SKU-specific proposals daily
			15	Flags operational exceptions	"The system also flags exceptions, like unusual demand spikes or potential shortages, which helps me focus my attention where it's actually needed."	Highlights anomalies to enable proactive intervention
			16	Improves service levels	"We've seen a noticeable improvement in service levels, fewer urgent shortages at construction sites."	Reduces stockouts through better anticipation
			17	Shifts daily workload	"In terms of workload, it's shifted rather than reduced. I spend less time on manual calculations and more time reviewing exceptions and coordinating with suppliers."	Frees routine tasks for higher-value activities
			18	Optimizes order quantities	"It also optimizes order quantities based on supplier constraints and transport efficiency."	Aligns orders with logistics to cut costs
			19	Enhances delivery visibility	"Internally we respond faster because we have better visibility."	Speeds internal response despite external lead times
			20	Reduces shipment frequency	"We're also consolidating shipments more effectively, which reduces the number of deliveries overall."	Lowers transport volume as a sustainability gain
Mikael Engberg	PostNord	Manager of Logistics Development	21	AI supports warehouse slotting and replenishment decisions	"We had a slightly different setup. It was more local in the warehouse, between the buffer zone and the picking zone."	The AI was mainly used to improve product placement and replenishment-related warehouse decisions
			22	ABC classification drives warehouse placement	"What it really did was put items into three categories: A, B, and C."	The system segmented SKUs by importance so the warehouse could propritize location quality
			23	Historical and forecast data support prediction	"We had a solution that looked at historical data" and "we needed something that could handle that."	The solution used past and expected demand patterns to decide how articles should be handled
			24	New SKUs required coding and prediction	"we needed something that could handle that" and "If there was a new article, we could look at a similar product from before."	The system helped classify new products even when no long sales history existed
			25	Productivity was the main performance goal	"For us, it was important to have good productivity when working with this."	The AI was valued because it improved warehouse efficiency and overall operational performance
			26	"Distance in picking" should be minimized	"It also helped keep the distance short when picking items. That was the main reason for us."	Better slotting reduced travel distance, which supported faster and more efficient picking
			27	AI reduced administrative workload	"It would mean more administrative time. That was what the system took away from us."	The tool reduced manual planning work and spreadsheet-based administration
			28	AI helped manage seasonality	"Before the season started, products needed to be sent out to all stores" and "When spring or another season came, the AI helped bring those products forward based on previous years."	The system adjusted placement and prioritization based on seasonal demand patterns
			29	C items needed proactive relocation	"the tricky part was mainly handling the articles that became C, because we had to move it to the C spot"	Slow-moving products required active intervention to avoid occupying prime warehouse locations
			30	Intergration with WMS was difficult	"we had a problem because of our WMS system. We needed to build bridges between the AI solution and the WMS"	Technical integration was a major implementation barrier
			31	System updates created problems	"Every time the WMS was updated, that created problems again"	The solution was vulnerable to changes in the surrounding IT environment
			32	The investment paid off	"when we looked at the time spent before and after implementation, we earned money from it"	The implementation delivered positive financial value
			33	Change management was a challenge	"It was also a challenge because people are used to working in a certain way, and then a new way comes in."	Adoption depended on convincing employees to trust the new approach
			34	Experienced staff were often more resistant	"people with a lot of experience tend to be more rigid"	Resistance to change was stronger among people who were used to established routines

Mikael Engberg	Sweco	Senior Logistics Consultant	35	AI is used frequently in everyday work	"I use it daily in my work for things like presentations and for figuring out what kind of questions to discuss with customers."	AI has become part of routine professional support work
			36	AI is still a support tool, not a decision-maker	"You need to look at whether it is actually reasonable."	Human judgment is still required to validate AI output
			37	Larger companies adopt AI more slowly	"Yes, I saw that earlier as well. Smaller or newer companies are more willing to explore it than bigger companies"	Bigger organizations move more cautiously because of complexity and risk
			38	Security and rules slow adoption	"big companies have a lot of security issues and other things to consider"	Governance and security requirments are major barriers to implementation
			39	Older organizations are more rigid	"In a big company, you often have different cities, countries, or departments, and that creates more structure and more barriers."	Long-established firms tend to have less flexibility and more internal fragmentation
			40	Smaller companies are more flexible	"New companies tend to adopt it earlier because they build it into the company from the start."	Smaller firms are often more adaptable and faster to try new tools
			41	AI adoption is still early	"I also think this is just the beginning of the AI era."	He sees AI as an early-stage technology with more adoption to come
Claudio Muñoz Percheron	L'Oreal	Core Supply Chain Business Expert	42	IT Collaboration is essential for process implementation	"Define the processes and work hand in hand with IT"	The interviewee bridges functional warehouse needs with technical implimentation by working closely with IT
			43	Core governance and arbitration is needed for proper alignment	"I had to defend that need with the core group"	Site-level needs must be justified in core governance before being added to the standard model
			44	Ongoing automation projects	"we are investing in automation in some sites"	The responsibility also includes adapting processes for warehouse automation projects
			45	Labor Management System is Important	"I am the current champion of this module"	The interviewee has ownership of the Labor Management System module as part of the WMS environment
			46	Warehouse Management System (WMS) as operational backbone	"the WMS is the transcations that are happening inside the site"	WMS is the main system for the warehouse transactions such as receiving, storing and dispatching
			47	AI model ecosystem is expanding across the organization	"we call it L'OrealGPT, but we also have Gemini and Claude"	AI use is becoming more diversified across the organization, not limited to one model
			48	AI training and democratization is happening across the company	"There is a democratization of AI use across the organization"	AI adoption is supported through broad training and organization-wide capability building
			49	SOP/ Manual Chatbots are used for operational knowledge	"We can load some of our SOPs or manuals into the system"	AI is used as a knowledge interface for operational procedures and internal manuals
			50	Order waving pre-analysis is used to predict execution problems	"the AI will analyze what kind of problems would happen if you wave these orders"	AI helps them identify problems before orders are realsed into execution
			51	Exception detection in waving is used to identify missing readiness	"if there is one SKU that does not have a picking location"	AI supports exception detection when operational readiness is incomplete
			52	Shipment alerts is used to flag late trucks and cutoff risks	"the truck is late"	AI is also use to warn users about transport timing risks and cutoffs
			53	Workforce distribution is used to support real-time labor allocation	"the AI can propose a distribution of your workforce in the different workshops"	AI supports real-time labor allocation in the warehouse
			54	Productivity improvement is a key expected benefit	"You gain productivity in terms of preparation management and truck filling"	AI is linked to more efficient preperation and shipping processes
			55	Business case building is necessary for implementation	"we are now understanding the benefits of AI"	The pilot is being used to justify the implementation through measurable values
			56	Cost justification is central to adoption decisions	"the license will increase the fee by X%"	AI adoption depends on whether the expected gains outweigh the addictional costs
			57	Benchmark site and constraints is a good example of warehouse efficiency under pressure	"they have constraints, but the organization they have put in place helps them overcome those constraints"	Strong warehouse performance can emerge from disciplined organization under constraints
			58	Slotting and SKU rotation is used to manage space and assortment constraints	"Every day, they change the slotting"	Daily slotting at the constrainted warehouse are used to manage large SKU assortments and limited space
			59	Bottom-up demand planning is the basis of replenishment flow	"there is a global process, and it is bottom-up"	Demand flows upward from countries before reaching production planning
			60	Demand planning and supply planning is the starting point of production planning	"All the countries have demand planning processes"	Demand planning is the starting point for supplu and replenishment decisions
			61	Factory production planning is driven by consolidated demand	"the factories receive the information through the process and then transforms it into a production plan"	Production is planned after demand is consolidated and arbitrated
			62	International Distribution Center (IDC) receiving role is mainly to receive and dispatch products	"the IDC follows the production of the factories"	Distribution centers mainly receive and dispatch what is produced upstream
			63	Quarantine and release flow is managed outside the factory to optimize production	"we have optimized the process so that quarantine does not happen in the factory"	Product release is organized so the factory focuses on production while the IDC handles blocked stock and release
			64	Requires Human Oversight	"There always needs to be a human in the middle chcecking the information coming from AI"	AI remains supportive and must be validated by people
			65	AI training is mandatory	"We have many trainings, even mandatory training, that employees have to follow to gain competencies in AI"	AI adoption is supported by compulsory training to build employee capability
			66	AI will be implemented this year	"this year, for sure, we will implement AI"	The interviewee expects AI to be implemented at L'Oreal this year after major steps were accomplished
			67	There is competitive pressure	"If we do not do it, our competitors will"	AI adoption is linked to competitive pressure and the risk of falling behind
			68	Forecasting software used already by planners	"we have some software tha we are using to estimate forecast based on data"	Demand planning already uses data-based forecasting software, likely as a foundation for AI use
			69	Beta testing with big tech	"they offer the possibility to some companies to become a beta tester. And on the AI effort, we pushed to be included in this group"	L'Oreal is participating as a beta tester in an AI pilot with a major technology provider, Manhattan
			70	Business value and cost pressure of AI	"we know that it works and adds value"	AI implementation is driven by expected value, but also constrained by added costs

Anonymous Interviewee 2	Anonymous Company 2	Regional Distribution Center Manager	71	AI-based forecasting and replenishment system is used	"We rely heavily on an AI-based forecasting and replenishment system that processes sales data from retailers, promotional plans, seasonality, and external factors like holidays or even weather patterns."	AI supports replenishment by combining internal sales data with external demand signals
			72	Daily replenishment proposals are generated by AI	"The system generates daily replenishment proposals for each SKU, which are automatically integrated into our ordering system."	AI directly supports SKU-level replenishment proposals and ordering processes
			73	Human role focuses on exception management	"My role is to review exceptions; like sudden demand spikes, anomalies, or supplier constraints and adjust when needed."	AI automates routine proposals while the manager focuses on exceptions and adjustments
			74	Human validation remains central	"So, instead of planning manually, I supervise and validate what the system suggests."	AI augments human planning rather than replacing decision authority
			75	AI predicts demand by SKU and location	"The system predicts demand per SKU, per location, and even adjusts for promotions or campaigns, which are very common in FMCG."	AI enables fine-grained replenishment planning across products and locations
			76	AI dynamically updates safety stock	"It also dynamically updates safety stock levels based on demand variability and supplier reliability."	AI supports adaptive inventory control by adjusting safety stock to uncertainty
			77	Manual management across thousands of SKUs is unrealistic	"That's something we couldn't realistically manage manually across thousands of SKUs."	AI helps manage complexity and scale in high-SKU warehouse environments
			78	Replenishment order automation increases capacity	"A large share of replenishment orders is generated automatically, which allows us to handle a much higher volume without increasing staff."	AI improves planning scalability by automating routine replenishment orders
			79	Service levels improved	"Service levels have improved significantly, especially during promotions where demand is harder to predict."	AI-supported planning improves availability under volatile promotional demand
			80	Excess inventory reduced	"At the same time, we've reduced excess inventory, particularly for slow-moving or seasonal items."	AI helps reduce overstock by improving alignment between inventory and demand
			81	AI helps reduce expiry risk	"The system helps us avoid overstocking products that risk expiring."	AI-supported replenishment can reduce waste risk for perishable goods
			82	Planners manage exceptions rather than routine decisions	"The difference is that we now focus more on managing exceptions rather than processing routine decisions."	AI changes the planner's work from routine processing to exception-based decision-making
			83	Lead times became more predictable	"Lead times are more predictable because we plan earlier and more accurately."	AI improves planning accuracy and predictability in replenishment processes
			84	Real-time sales data supports response to demand changes	"The system continuously updates forecasts based on real-time sales data, so we can react quickly to changes in demand."	Real-time data improves responsiveness to demand fluctuations
			85	AI simulates supplier delay impacts	"For supplier delays, the system simulates the impact on inventory and suggests mitigation actions, like increasing orders from alternative suppliers or reallocating stock between warehouses."	AI supports resilience by modelling disruption impacts and proposing mitigation actions
			86	Extreme disruptions still require manual intervention	"However, in extreme situations like sudden supply chain disruptions, we still rely on manual intervention and experience."	AI supports disruption response but does not replace human judgment in exceptional cases
			87	AI reduces waste for perishables	"There's been a clear impact on waste reduction, especially for perishable goods."	AI-supported replenishment can contribute to sustainability by reducing product waste
			88	Transport efficiency improved	"We've also improved transport efficiency by consolidating shipments and reducing urgent replenishment orders."	AI-supported planning can reduce urgent shipments and improve logistics efficiency
			89	Cross-functional collaboration increased	"Collaboration has also increased, we work more closely with sales and marketing teams because their input, like promotions, directly affects the AI forecasts."	AI-supported replenishment depends on integration between warehouse planning and commercial functions
			90	Data quality and integration affect usefulness	"Forecast accuracy depends heavily on data quality and integration."	Data quality is a key technical condition for effective AI-supported replenishment
			91	User trust is uneven	"Some planners initially relied too much on the system, while others didn't trust it enough."	Effective AI use requires balanced trust and appropriate reliance
			92	Explainability would improve trust	"In terms of improvements, better explainability would help understanding why the system makes certain recommendations would make it easier to trust and use effectively."	Explainability is important for improving planner confidence and effective AI adoption

Karim Nouria	Sics AI	Founder	93	Robot-controlling AI is used for order picking	"it controls robots in warehouses and industrial settings"	The interviewee's AI is applied directly to warehouse order picking through robot control
			94	AI is robot agnostic	"It is robot-agnostic, so it can control any type of robot and use any type of sensor, and one AI controls all the robots."	The system is designed to work across robot types rather than being tied down to one machine
			95	AI has strong generalization ability	"Once it learns something, it can apply that in other situations."	The AI can transfer learning across different situations
			96	AI is deployed in an industrial robot	"We have our first AI deployed in an industrial robot at a third-party site in Helsingborg, with amazing performance and capabilities."	The technology is already being used in a real industrial setting
			97	AI plans the individual pick	"It controls the individual pick."	The AI works at the level of individual picking execution
			98	AI is replacing manual pickers	"Primarily, we are replacing manual pickers in the warehouse."	The AI is changing labor roles in the warehouse
			99	AI is four times faster than a human	"We pick about four times faster than a human, depending on the objects."	The system significantly increases picking speed
			100	AI can work 24/7	"We can work 24/7"	The robots can continuously operate without human limits, something that is impossible with human-centered warehouses
			101	AI increases warehouse efficiency	"So it increases warehouse efficency"	Since AI is faster than humans and can work 24/7 independently, the system improves overall warehouse performance
			102	AI provides excellent transparency	"In addition, you get excellent transparency"	The system improves visibility into warehouse operations
			103	AI gives better forecasting input	"With this increased supply chain transparency, [...] better forecasting and prediction ability."	The operational data can support forecasting and predictions
			104	AI follows a different domain from forecasting AI	"I think existing AI technology can be used for that, and it is already used in some ways. But that is a different domain from where we are right now."	Robotics AI is distinct from planning and forecasting AI already in the market
			105	AI learns by interaction and feedback	"It learns more like a child"	The model improves through experience and feedback
			106	AI can save money on investment	"On the other hand, with more efficient picking, you do not need as much storage capacity anymore. So you can save money on investment and use that money for investments in other parts of the supply chain. "	AI-enabled picking efficiency reduces the need for excess storage capacity, which lowers capital requirements and allows resources to be redirected to other supply chain investments.
			107	AI contributes to sustainability	"Yes. As long as you provide high efficiency, you use resources and energy more efficiently, so of course it contributes to sustainability. "	AI contributes to sustainability indirectly by improving efficiency and reducing resource and energy consumption in warehouse operations.
			108	AI is moving toward other applications	"What we are developing is a more general AI. We are starting with this robot application, but we will move to other types of applications."	The current use case is a starting point for broader adoption
			109	AI is moving toward robot orchestration	"Right now, we only control individual industrial robots. In the future, we will have robot orchestration functionality, but we do not have that yet. "	There are long-term plans to coordinate the control of multiple robots
			110	AI is based on federated learning	"If one robot learns something on the job, the other robots also get access to that knowledge. It becomes a knowledge explosion within the system."	The statement suggests a fragmented learning architecture, where each robot accumulates experience independently rather than contributing to a collective shared model.

Bernt Hylen	Consafe	Product Owner AI and Digital Twin	111	Machine learning is used	"Most features in Astro use machine learning"	Demonstrates ML as core enabler for adaptive features, making Astro a data-driven system optimizing dynamic warehouse operations through predictive insights
			112	Different replenishment types are offered	"three main types: order-driven, low watermark, and predictive replenishment."	Shows flexible replenishment strategies made to order patterns, enabling warehouses to balance inventory precision with operational responsiveness
			113	Assignments are triggered	"Order-driven replenishment triggers assignments when you release picks for known orders"	Shows reactive automation linking picking to replenishment, minimizing stockouts and streamlining order fulfillment workflows
			114	Assignments are created	"Low watermark replenishment creates assignments when inventory falls below a set level in pick locations"	Illustrates safety-net mechanism maintaining pick-face availability, critical for high-velocity picking without manual intervention
			115	Solution is also suited for building industry	"Predictive replenishment suits businesses like the building industry where orders arrive late in the afternoon but must be ready early next morning"	Highlights sector-specific adaptation for late-afternoon orders, transforming chaotic demand into proactive morning replenishment for next-day delivery
			116	Demand is forecasted	"It forecasts demand using historical patterns, so warehouses replenish in the morning when staff is available and focuses on picking later"	Enables resource allocation via pattern recognition, reducing peak-hour chaos and aligning labor with anticipated needs
			117	Workload is spread	"This spreads workload, cuts overtime costs, and improves work-life balance"	Promotes sustainable operations with cost savings, employee well-being and workforce allocation
			118	Metrics are improved	"These methods improve service levels, inventory control, lead times, and workload differently based on business needs"	Shows overall performance improvements by raising service levels, cutting inventory costs, and speeding up order processing
			119	Processes are handled	"Astro WMS handles all warehouse processes from inbound to outbound, including picking, packing, and value-added services like labeling or assembly"	Astro is considered as an orchestrator, unifying fragmented activities into seamless flow
			120	ERP is integrated	"It integrates with ERP systems for articles and orders"	WMS execution complements ERP planning, fostering enterprise-wide synchronization without redundant forecasting
			121	Manual areas in the warehouses are supported as well	"It supports both manual and automated warehouses"	Supports hybrid scalability, bridging manual operations to modern automation for phased digital maturity
			122	Automation levels are supported	"integrating at levels from full control (white box) to minimal (black box)"	Works with all automation types (white: Astro WMS completely controls the automation equipment/black box: The automation equipment controls itself, Astro just sends pallets/ Gray: they share control) so customers can use any vendor's hardware.
			123	Storage is optimized by mapping	"Warehouse mapping optimizes storage by frequency and associations while respecting constraints like heavy items first to avoid damage"	Slashes travel time while respecting physical constraints like fragility
			124	AI mapping is developed for future	"We're developing AI for automated mapping, expected in 1-2 years"	Astro will be able to automate complex multi-constraint slotting
			125	Associations are already included	"Current AI includes article associations for optimal placement"	Uses "customers who bought this also bought that" logic to put related items together, so pickers walk less and work faster
			126	Traveling problems are solved by routes optimization	"Route optimization solves the traveling problem for shortest pick paths, but requires distance mapping"	Uses "shortest route" math to plan pick paths (needs good warehouse map first) so workers walk (or drive) less in big warehouses with many products
			127	Orders are combined through batching	"Batching combines orders to fill pallets efficiently"	Groups multiple orders onto one pallet to fill trucks better and plan waves so urgent orders ship first
			128	Forklifts availability are optimized when empty	"and we optimize empty forklift moves"	Plans empty forklift routes (like replenishment paths) to cut wasted trips in heavy storage areas
			129	Adoption seems harder than expected to some	"Users struggle to implement new features due to time constraints and resistance to change, often believing their operations are unique"	Busy warehouse teams lack time to use new features, missing out on their benefits
			130	Workforce allocation is predicted by AI	"Workforce prediction is another AI area"	Predicts exact worker needs by area and shift, using sales forecasts to adjust staffing quickly when order volumes change
			131	Forecasts are enhancing AI capabilities	"enhanced by external sales forecasts from partners"	Links warehouse system to sales predictions (sales/new items) so AI plans better for surprises
			132	Job threat is posed by automation	"Automation, not planning tools, poses the biggest job threat by replacing manual labor"	Robots replace bulk manual jobs, AI helps planners work smarter, overall fewer workers needed due to efficiency, not less brainpower

Anonymous Interviewee 3	H&M	Warehouse Responsible	133	Reliance on human expertise instead of AI	"We don't use AI; we use really smart persons"	The organization currently relies on skilled employees and experiential judgment rather than AI tools for decision-making. This highlights that AI adoption in warehouse planning is uneven and that human capability remains a key resource in replenishment decisions.
Julia Lindblom	Göteborgs Posten	Investigative Journalist	134	Exploitation of Data Labelers	"AI revolution's workers traumatized by gross images"	Labelers face psychological harm from labeling violent/graphic data for AI systems
			135	Raises ethical concerns	"working conditions of annotators worldwide," rejecting "a little work is better than none."	Critiques global labor abuses enabling AI profits, demanding living wages
			136	Questions master data collection	Ongoing investigation into annotators feeding data to AI products via human feedback	Highlights unethical, low-wage data pipelines as hidden cost of AI efficiency
			137	Urges social responsibility	Covers tech giants' failure to protect workers	AI adoption must prioritize fair conditions over unchecked growth
Filip Höcker	Consafe	Sales Executive	138	Small companies embrace AI implementation eagerly	Small-sized companies are eager to implement AI in their processes more than the bigger ones	Smaller firms drive faster AI uptake due to agility and lower inertia, contrasting larger ones' caution
			139	Data volume an important aspect for service providers	Data volume coming from more mature companies are more valuable to us as service providers, than the ones from the smaller companies	Service providers gain better insights and models from large firms' abundant, structured data over small firms' limited volumes
Anonymous Interviewee 4	Unilever - Dubai	Logistics Lead Responsible	140	Absence of AI tools	"Currently, no. We don't own the premises [...] We're not yet at the stage of implementing AI there"	Unilever UAE has not yet implemented any AI systems within its warehouse operations. This reflects a pre-adoption stage where awareness of AI's potential exists, but structural and contractual constraints have prevented any formal deployment
			141	Manual decision-making	"Manually gathering and interpreting data to make decisions is time-consuming and creates inefficiencies for the team."	Manual work exposes the operation to slower response times, human error, and resource inefficiency, highlighting the performance gap that AI could address
			142	Primitive tools in use	"I'm currently using some basic tools to bring data together and create some visibility for decision-making."	This represents an informal, individual-driven workaround rather than a systematic organizational solution, indicating a gap between operational needs and available technology
			143	Route optimization enhancement	"The first thing that comes to mind is route optimization, that's something heavily driven by data and well-suited for AI."	This suggests that last-mile delivery efficiency is perceived as the clearest value proposition for AI adoption in outbound logistics
			144	Internal logistics and warehouse enhancement	"The second area is internal logistics: analyzing data to facilitate key decisions around warehouse enhancements."	AI is seen as a tool to improve internal warehouse activities through data-driven decision-making
			145	Cost control and waste reduction	"Route planning and optimization could also significantly increase our effectiveness and efficiency in deliveries while reducing cost and waste."	It reflects a financially motivated view of AI adoption, where return on investment is central to justifying implementation
			146	Data visibility enabling faster decisions	"Having AI bring that data together and provide visibility would be really valuable, not just for operational decisions but also for cost control."	The current lack of visibility is a bottleneck that slows decision-making across all operational areas, and AI is viewed as the mechanism to overcome this through better data visibility
			147	3PL dependency	"any specific software, programs, or infrastructure requirements would need to be negotiated and agreed upon."	This structural dependency significantly delays and complicates adoption, as decisions cannot be made unilaterally by Unilever
			148	Cost-benefit justification required	"It would need to be justified through a cost-benefit analysis."	It reflects a risk-averse organizational culture where financial justification is a prerequisite for change
			149	Fear of job replacement	"If you're speaking to a warehouse team member who has decades of experience doing things a specific way, they're more likely to feel threatened and resistant."	The fear of experiencied workers is rooted not only in the nature of AI itself but also in broader anxieties about technological displacement, representing a significant human barrier to adoption
			150	AI perception shaped by background	"The level of existing knowledge greatly influences how threatened someone feels."	An individual's educational background, professional experience, and digital literacy all shape how they perceive and respond to AI.
			151	Investment in people's digital capabilities	"There's also investment going into building people's digital and AI capabilities, which could support larger investments down the line."	Investing in digital skills and AI literacy is seen as a foundational step before larger technological investments can be made effectively and embraced organizationally

8.3 Gioia Matrix

This Gioia Matrix represents one of the five aggregate dimensions and its corresponding second-order themes and first-order concepts. It is a representation of the process that has been followed. Refer back to the coding table (Appendix 8.2) to read more about the first-order concepts mentioned in the matrix below.

